Foundations **of** Computation

COMPUTATIONAL COMPLEXITY

GEOMETRIC VERDICTS ON P VERSUS NP

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Why the Structural Verdict on P versus NP Is the Only One Left Standing

Mathematics Proves That Mathematics Cannot Rule Here

Riemann's Line Is Handed Over by a Symmetry That Never Mentions a Zero; P versus NP's Target Set Is the Question Itself

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Structural evidence is ordinarily subordinate to proof: *the evidence points one way, but only a proof settles it.* That subordination rests on a premise nobody states because nobody needs to, namely that the court of final appeal is open. This paper proves it closed, and then asks what stands. The closure. Stripped of five interpretive layers, P versus NP is a Π^0_2 tail-statement about a machine Levin constructed in 1973. The Riemann hypothesis is a residence claim on a set a symmetry hands over: the critical line is the fixed set of $s \mapsto 1 - \bar{s}$, the functional equation composed with Schwarz reflection, and neither factor mentions a zero. P versus NP has no such symmetry. By Cook-Levin its target set is P, the object the conjecture asks about, and you cannot localize onto the question. Its most natural symmetry, complementation, hands over nothing: fixed set empty, dimension zero, sixteen languages in eight two-cycles. The string therefore takes [$\emptyset_0$], the grounded-sealed halt. This is not the ordinary open verdict, where the file is incomplete and the repair is named; it is the finding that the file is complete and no standpoint the verdict side occupies reaches the string. The halt is sealed, terminal against method, reframing, and instrument. A proof of either direction is not offered here and is not claimed. Fifteen barriers in three classes support it, sorted by quantifier rather than strength: technique-indexed walls arriving after the families they block at lags of ten, nine, eighteen, fifteen, and two years; type-indexed walls that quantify over the structure any technique must engage and therefore bind the unborn identically with the born; and one closure result proving the catalogue cannot certify its own completeness. What stands. Two further faces seal at Type T without deducing anything: the kinetic field, a completed physical event bounded by the thermodynamic arrow and answer-independent by proof, its rows taking identical values under both truth values of the separation so the mutual information is exactly zero; and the geometric shape, silent on the truth-sign by an identity. Neither seal deduces the string, and the reason is a theorem: the single premise between them is logically identical to the conjecture by Cook-Levin and admits no physical discharge. The direction comes from elsewhere and is checkable: at the time register every decided cousin separates, unconditionally at computability and by Paul-Pippenger-Szemerédi-Trotter at multitape linear time, while the cousins that collapse are space cousins one register away. The inversion. With the deductive layer proven unreachable, the structural determination is not provisional pending a ruling that cannot come. It is the only determination available at any standpoint that exists. Its grade is non-deductive and it never becomes a theorem. The forfeit is itself a theorem, and every leg of it is classical and cited: mathematics is what proves that mathematics cannot rule here. No new mathematics is introduced anywhere, which is the result's armor rather than its limit, since there is no invented object to attack and the entire surface is an arrangement that is printed and executable.

KEYWORDS P versus NP, barrier theorems, relativization, natural proofs, algebrization, fixed-point-free involutions, arithmetic hierarchy, structural diagnosis, Riemann hypothesis

1 · The Question and the Shape of Its Resistance

The P versus NP question asks whether every decision problem whose solutions are verifiable in polynomial time is also solvable in polynomial time. Formalized by Cook in 1971 and independently by Levin in 1973, extended across the combinatorial landscape by Karp in 1972, and designated a Millennium Prize Problem in 2000, it has absorbed five decades of concentrated effort without yielding. This paper does not resolve it. It does something the record now supports and that has not been done: it determines the structure of the resistance itself, and it argues that the structure is not what the standard reading assumes.

The standard reading is that the problem is hard, that the known techniques have been tried and found insufficient, and that a sufficiently novel technique may yet close it. This reading is not wrong so much as underdetermined. It treats the openness as an epistemic state, a fact about what has been found, and it therefore licenses the inference that more searching may change it. There is a second reading available, and the difference between the two is the subject of this paper. On the second reading the openness is a structural fact about the terrain the question inhabits, in the precise sense that it survives a complete diagnosis rather than awaiting one.

Distinguishing these two readings requires more than rhetoric. It requires exhibiting the diagnosis and showing it complete. We do that in four movements, and each is checkable independently of the others.

The first movement is an intake operation, and it is prior to everything else. The famous question, whether computers can solve hard problems quickly, and the formal string, a Π^0_2 sentence quantifying over machines and exponents, are routinely treated as one object wearing two names. They are not. Section 3 exhibits the divergence, shows that the identification is unsupported in every world and contradicted in one that nothing currently excludes, and shows that its natural repair route is itself blocked by a theorem. Five interpretive layers strip away. What survives is a tail-statement about an object already constructed and sitting in the literature since 1973.

The second movement is a computation, and it is elementary. Section 4 computes the fixed-point structure of the symmetry the surviving string most naturally suggests, and states the criterion under which a symmetry is native to a conjecture at all. The result is that complementation on the language space has no fixed points, so the associated linear map on the indicator space is minus the identity: every eigenvalue is minus one and the fixed subspace has dimension zero. We contrast this against the Riemann hypothesis, whose native involution is the functional equation composed with Schwarz reflection, $s \mapsto 1 - \bar{s}$, and whose fixed set is the critical line, a nonempty determinate object of dimension one. The functional equation **alone** fixes only the point 1/2 and carries complementation's signature exactly, so Riemann's line is bought with a second symmetry rather than handed over free. The difference between the two questions is the **conjecture's form**: one resides on a set a symmetry delivers, the other on P, the object it asks about. The one measured bit of dimension, one against zero, is not that difference but what **sorts** the two opennesses into different species, and Section 4.4 is where the distinction is held exactly.

The third movement is an assembly, and it is the paper's largest section. Section 5 collects the complete barrier record. The literature usually names three barriers. There are more, and more importantly they are not all of one kind. We sort them by a property that has not, to our knowledge, been made the organizing axis before: their temporal structure relative to the techniques they block. Some barriers chase; they arrive after the technique class is developed, and the record of arrival lags is stark. Others do not chase, because they do not quantify over techniques at all. This distinction is what carries the paper's central claim, and Section 10 makes the argument in full.

The fourth movement is a pair of theorems, and they are short. Section 7 shows that the single premise separating any bound on physical resources from the class separation is logically identical to the conjecture. Section 8 shows that no physical argument discharges that premise without proving the conjecture, and that the counting-type bounds specifically are blocked by relativization. Together they measure the gap between a physical fact and the abstract question, and the measurement is exact: the gap is one premise wide, and the premise is the conjecture.

Sections 9 and 10 state the composite verdict and, more importantly, state precisely what it licenses and what it does not. Section 10 is the paper's argumentative core: it distinguishes a barrier that will be defeated by a technique not yet invented from a barrier that will not, gives the criterion, and shows why the criterion is not a prediction about the future but a fact about types.

1.1 · The Composite, Stated First

The paper's result is one table, and everything after it is the derivation. This is the only table in the paper that carries the verdict glyphs; every other statement is in prose, by design, so that the tokens mean something where they appear.

Table 1 | The composite verdict on P versus NP. Three faces on the proposition, decided separately and never conflated. The tuple does not compress, and Section 9.5 is why.

Layer / Face	Token	Typing / Grade	Mechanism and definition
1 · The kinetic field	[ӄ] ACTUALIZED INVARIANT	Type T (physics)	The verify-generate asymmetry read as a completed physical event. Bounded by the thermodynamic arrow and the actuation cost. **Answer-independent.**
2 · The geometric shape	[ӄ T] SEALED	Type T (algebraic)	The problem's **most natural symmetry**, complementation, native to a different question by §4.3's criterion, is **fixed-point-free**. Eigenvalues all minus one. Reflexive membership is timeless. A shape fact, silent on the string's truth-sign.
3 · The formal truth-string	[$\varnothing_0$] **GROUNDED-SEALED HALT**	structural, on Type-T legs / premise cap	Issued on the stripped Π^0_2 string, and **not the ordinary open verdict**. Per §2.2 the ordinary verdict reports an incomplete file and names the repair; this one reports a **complete file**. The halt is **earned and sealed**, not surveyed: the string is determinate on $\mathbb{N}$-truth at the premise cap, no independence proof exists, the obstruction is proven rather than reported, and no standpoint the verdict side occupies reaches it. Fixed-subspace dimension **zero** is §2.2's one measured quantity, and it is what sorts this from Riemann's [Ξ_0], the sibling refinement where the file is likewise complete but the obstruction is blindness in the reader before a line that exists: **the halt lives in the walls, not in the reader.** What is sealed is the halt. What is not claimed, here or anywhere in this paper, is a proof of either direction.

1.2 · The Thesis: The Court of Final Appeal Is Provably Closed

The paper has one claim and it is not about P versus NP. It is about **what stands when the deductive route is proven unreachable**, and it is the reason Row 3 is worth having.

The standard hierarchy, and its hidden premise. Structural, physical, and convergent-evidence considerations are ordinarily subordinate to proof. Everyone in the field knows the form: *the evidence points one way, but only a proof settles it.* The subordination is correct and it is also load-bearing on a premise nobody states, because nobody needs to. The premise is that **the court of final appeal is open for business.** Structural evidence is provisional precisely because a proof might arrive and overrule it. Take away the possibility of the proof arriving and the word *provisional* loses its referent, because there is nothing left for the evidence to be provisional **to**.

Row 3 takes it away. Not by observing that no proof has arrived in fifty-five years, which is an epistemic report and licenses nothing. By establishing that **from every standpoint the verdict side occupies, no proof is reachable**, on a conjecture whose target set is P, the object in question, per §4.3, with its most natural symmetry's fixed subspace measured at dimension zero, sealed on the conditional at the declared premise cap, terminal against method, against reframing, and against instrument. The court is not in recess. It is structurally closed, and the closure is itself a mathematical result.

So the structural determination is not the junior partner waiting on the senior's ruling. It is the only determination available at any standpoint that exists. That is what the halt buys, and it is the whole point of paying for it. This is the inversion the paper is for: the layer that is normally supreme has provably forfeited, and the layer that is normally provisional is left holding the only verdict there is.

And the supremacy is not a victory over mathematics. It is a gift from it. The forfeit is a theorem. Complementation's fixed-point-freeness is one line and is classical. The barrier record is Baker-Gill-Solovay, Razborov-Rudich, Aaronson-Wigderson, and the rest, every one of them somebody else's theorem. The quantifier classification is classical. **Mathematics is what proves that mathematics cannot rule here**, and it hands the ruling to the geometry on its way out. Section 1.3 is why that matters more than it sounds: the transfer costs nothing, because nothing was invented to effect it.

What the geometry then delivers, and at what grade. Rows 1 and 2 seal at Type T without deducing anything. Three warrant streams, populated independently and gated on a deletion test and an isolation test, either enclose a nonzero volume or they do not, and **there is no probability in that fact**: the Gram determinant is strictly positive exactly when the streams are linearly independent, by Cauchy-Binet; the margin is the distance to degeneracy, by Eckart-Young and Weyl; the verdict is invariant under every rotation, so the labels are gauge and the geometry is the content. A convergence that encloses volume is an **absolute geometric necessity given its inputs**.

Where the direction comes from, stated exactly, because this is where a reader will look for the sleight of hand. It does not come from Rows 1 or 2. Row 1 is **answer-independent**, at a mutual information of exactly zero by §9.3, the recorded estimator sitting at its own noise floor of 7.19×10^{-5} with the separation, and Row 2 is **silent on the truth-sign** by an identity. Their own mechanism columns say so and the paper does not walk it back. The direction comes from a third place, and it is checkable: **at the time register, every decided cousin separates.** Computability separates, unconditionally. Multitape linear time separates, by Paul-Pippenger-Szemerédi-Trotter. The cousins that collapse, polynomial space by Savitch and log space by Immerman-Szelepcsényi, are **space** cousins, and Section 3.6 records honestly that they cut against the general grain argument and permanently weaken it. But the open question is a **time** question, and the time register is two for two. Add the barrier record, in which every technique that came close was walled by its own invariance signature, and the physical arrow, under which the trajectory that finds is thermodynamically distinct from the trajectory that checks. **That is the convergent evidence, and it points at separation.**

Its grade is non-deductive and it never becomes a theorem. Held outside the theorem register permanently, revisable by exactly one thing: a supplied proof, which would **replace** the determination and not correct it, because the determination never claimed to be what the proof would be. And no such proof is reachable from where anyone stands, which is Row 3, which is the whole argument closing on itself.

So: the deductive layer forfeits by a theorem; the geometric layer is left holding the only verdict available at any standpoint; the verdict points at separation on convergent evidence at non-deductive grade; and not one line of mathematics was invented anywhere in the chain. **That is the paper, and Section 1.3 is why the last clause is the strongest one in it.**

1.3 · Zero Invented Mathematics, and Why That Is the Armor

The paper introduces no new mathematics. Not one lemma, not one construction, not one definition doing load-bearing work that was not already in the literature. Complementation's fixed-point-freeness is one line and is as old as set theory. The barrier theorems are Baker-Gill-Solovay 1975, Razborov-Rudich 1994, Aaronson-Wigderson 2008, and the rest, each cited to its source. The completeness identity is Cook-Levin 1971. The determinant facts are Cauchy-Binet, Hadamard, Eckart-Young, and Weyl. The tower results are Turing 1939 and Feferman 1962. The register split is Savitch, Immerman-Szelepcsényi, and Paul-Pippenger-Szemerédi-Trotter. **Every leg is classical and every leg is somebody else's.**

This is routinely read as modesty. It is not modesty. **It is the load-bearing structural property of the result, and it is what makes the seals hard to attack.**

Consider the alternative. A paper that invents mathematics to settle a question of this size hands its reader an obvious target: the invention. The invention is the newest, least-examined, least-replicated object in the argument, and every reviewer knows to go there first, and they are right to. The invention is the weakest link precisely because it is the newest link, and no amount of care changes that ordering.

A paper that invents nothing has no such link. The seals of Table 1 rest on results that have been attacked for decades by everyone who cared to and are still standing. There is no fresh object to break. A reader who wants to defeat Row 2 must defeat the claim that no language equals its own complement. A reader who wants to defeat Row 3's wall record must defeat relativization or natural proofs. **Attacking the legs of this paper is not attacking this paper; it is attacking the field.**

What *is* attackable is the **arrangement**: the encoding, the stream assignment, the deletion and isolation tests, the two ranks, the sorting of the barriers by quantifier, the routing of the gates. That is where every real objection to this paper lives, and it is exactly where we have put the reader: the tests are executable and unified: every computed number in the paper is produced by one battery, at one seed, in one pass, and the run is certified by a single digest recorded in Appendix H, so *the code reproduces bit for bit* is not a promise but a checkable hash. The two ranks are **argued separately and the arguments are printed** rather than read off a determinant, and the barrier sort is tabulated with its criterion. One thing is not put in front of the reader and we say so rather than let it be found: **the warrant encoding itself is never printed**, no $\det(R)$ is computed on it, and Appendix F's exhibits run on synthetic input because the identities they demonstrate hold for arbitrary input. A reader who wants to attack the encoding is entitled to the complaint that there is nothing to attack, and that complaint is correct. **The attack surface has been relocated from an invention nobody can check to an arrangement anybody can.**

So the zero is not a confession and it is not a limit on the claim. It is the reason the claim can be made at all. **A permanent structural seal on the formal string of a millennium problem, reached with no new mathematics, is a stronger object than the same seal reached with new mathematics would be**, because the arrangement is checkable in an afternoon by anyone who knows the classical results, and the invention would not be.

The discipline has exactly one exit, and it is gated on what the instrument cannot forge. A claim of authored new mathematics requires an external, independent, reproducible, adversarially audited witness: a proof-assistant certificate or a referee, produced by someone who is not the claimant. No internal reasoning, however clean, opens that gate. This paper does not approach it and does not want to.

1.4 · What Is Claimed and What Is Not

We are explicit at the outset in both directions, because the two failure modes are symmetric and a paper that guards only against one of them has already committed the other.

What is settled is settled. The bridge premise is the conjecture, by a theorem. It has no physical discharge, by a theorem. The most natural symmetry is fixed-point-free with a fixed subspace of dimension zero, by computation at machine precision. The squared invariant cannot carry a truth-sign, by an identity. The barrier record assembles into a statement of zero access from every verdict-side standpoint, with the deciding input constitutively supplied. And the openness of the stripped string is located in the terrain rather than in the search, on a diagnostic that is complete. These are not offered tentatively and they are not softened below. Section 12 states them together at the weight they earn.

What is not claimed is not claimed. No proof of P ≠ NP is offered. No proof of P = NP is offered. No independence result is offered; no such theorem exists for this problem, and we do not claim one. The Clay Institute criteria are not met and are not approached. The conjecture remains open in the plain sense that no proof of either direction exists.

What is offered is a structural diagnosis with the following four components, each stated at its own warrant. A theorem: the bridge premise is the conjecture (Section 7, elementary from Cook-Levin). A theorem: that premise has no physical discharge (Section 8). A computation: the symmetry the stripped string most naturally suggests is fixed-point-free, exhibited at machine precision, and no symmetry is native to it in §4.3's sense (Section 4). An assembly: the barrier record in three classes with the temporal law of each class (Section 5). The conclusions we draw from these are structural rather than deductive, and every structural conclusion is marked as such at the point of use.

The distinction between a diagnosis and a resolution is one this paper insists on at every boundary. A complete diagnosis of why an instrument does not reach a target is not a determination of where the target is. We show the terrain. We do not cross it.

2 · Method: Three Independent Verification Instruments

The analysis runs on a verification architecture whose relevant machinery we state here in standard terms, without framework-internal vocabulary. Readers interested only in the mathematics may read Sections 3, 4, 5, 7, and 8 and skip this section entirely; nothing in those sections depends on the apparatus described here.

2.1 · Three Warrants, Independently Populated

Any claim under audit is decomposed into three warrant streams that must be populated independently and must not be reconstructible from one another. The **structural** stream carries formal and derivational support: theorems, reductions, lower bounds, proven barriers. The **empirical** stream carries physical and measurement support: observed scaling, resource bounds, thermodynamic and information-theoretic floors. The **registrational** stream carries support concerning the relation between an observer's conviction and that observer's capacity to act on it: the structure of interactive proof systems is the canonical source.

Two independence tests gate admission. The **deletion test** removes each stream and asks whether the support it carried is recoverable from the remaining two; if it is, the stream was not independent and is reclassified. The **isolation test** requires that each stream be statable in vocabulary that cannot reconstruct the others; a stream whose content back-translates into another stream's vocabulary is reassigned to the stream it actually belongs to.

The isolation test is not a formality. It is the operation that keeps a paper of this kind from counting one piece of evidence three times under three names, which is the standard failure of convergence arguments.

2.2 · Three Verdicts, and Two Refinements Inside the Open One

The verdict economy is three-valued and stays three-valued. A claim is **sealed** when the three streams converge non-degenerately and every admissibility condition holds; **broken**, with a named mechanism, when a condition fails or when no stream supports it; or **open**, with a named violation, otherwise.

There is no fourth verdict. Two refinements live *inside* the open verdict, and they are the subject of Section 9. Both mark a situation where the file is complete and the reader nonetheless cannot read the answer, as against the ordinary open verdict where the file is incomplete and the repair is named. They differ from each other by exactly one measured quantity, computed in Section 4.

2.3 · The Target-Indexed Evidence Criterion

A warrant stream counts for a proposition only if its likelihood ratio favors that proposition, that is, only if the stream is more probable given the proposition than given its negation. A stream present in all relevant worlds carries a likelihood ratio of exactly one and does not count for that proposition, though it may count for a different one it does favor.

This criterion is indexed to the target and the indexing is load-bearing. The same body of evidence yields different counts against different propositions, and Section 9.3 shows that the two propositions most often conflated in this literature take different counts from the identical evidence. Failing to index the criterion is the error that lets a physical observation appear to support an abstract separation it is in fact silent about.

2.4 · The Mass Requirement on Explanations

A factor offered as a common cause of an observed convergence is admissible only if it carries a measurable signature, an energy or entropy cost associated with the factor. A factor that explains no measured variance cannot be the common cause of an observed agreement. The restriction runs symmetrically: a factor without measurable signature can neither establish a claim nor break an established one. This is why, in Section 5, a history of failed attempts is not admitted as a barrier: failed attempts are a record of what was tried, not a theorem about what can be tried.

3 · The Stripped Object: Separating the Famous Question from the Arithmetic String

Before any audit runs, the object under audit must be well formed. This section establishes that the object usually audited is not, and identifies the one that is. The finding is prior to everything downstream: if the famous formulation and the formal string are distinct objects, then five decades of argument about "the" problem has been argument about two.

The claim under examination is the identification itself: that the question *can computers solve hard problems quickly* and the sentence *for all machines and all exponents there exists a failing input* are one proposition. The identification was installed by the Cobham-Edmonds convention and has been repeated since. Three witnesses show it fails, and they fail it at three different grades.

3.1 · Witness One: The Divergence World

Impagliazzo's five-worlds partition names a world, Heuristica, in which the separation holds and yet every problem in NP is easy on average. In Heuristica the arithmetic string is **true** and the famous meaning is **void**: no machine decides the NP-complete language in polynomial time in the worst case, and simultaneously nothing anyone actually wants to compute is hard. The identification asserts that these two cannot come apart. Heuristica is a world in which they do, and nothing currently excludes it.

This witness is conditional-grade: it rests on the five-worlds partition being a genuine partition rather than a heuristic taxonomy, and on Heuristica being consistent. Both are standard. Neither is proven. The witness therefore shows the identification is *contradicted in an unrefuted world*, which is weaker than refutation and stronger than doubt.

3.2 · Witness Two: The Missing Bridge and Its Blocked Repair

The natural repair is a theorem carrying worst-case hardness to average-case hardness for NP, which would close Heuristica and restore the identification. No such theorem exists. Worse for the repair, its most natural route is walled: Bogdanov and Trevisan proved in 2006 that a non-adaptive worst-case-to-average-case reduction for NP collapses the polynomial hierarchy.

This is theorem-grade conditional, and its scope is exact. It bars the non-adaptive lane. Adaptive reductions are not barred by it and remain a live target. But the identification is not merely unproven; along its most natural repair road there is a theorem in the way.

3.3 · Witness Three: The Intake Test, Executed

The third witness is mechanical and it is the one a reader can run. Consider the two candidate populations for the slot the name claims to fill.

Population A, the famous object, offers this vocabulary: *feasible*, *fast*, *in practice*, *real machines*, *reasonable constants*, *tractable*. **Population B**, the stripped string, offers this: *machines*, *exponents*, *inputs*, *asymptotic tail*.

The slot requires a well-formed arithmetic sentence. Population B parses into it. Population A does not: its vocabulary is about engineering practice on bounded inputs with bounded constants, and there is no arithmetic sentence any of its terms denote. Population A is therefore **refused at intake**, not refuted. Population B is **admitted**: it is a well-formed sentence of arithmetic, and Section 3.4 classifies it.

One name, two objects, and only one of them parses at the register the name claims. That is the contradiction, and it is executable rather than argued.

3.4 · The Five Layers, Stripped

Five interpretive layers sit between the famous question and the arithmetic string, and each has an exact type.

Layer one, the feasibility identification. The Cobham-Edmonds convention identifies polynomial time with feasibility. The identification fails in both directions and the counterexamples are standard: a running time of n^{1000} lies in P and is infeasible on any input anyone will ever present; a running time of $2^{(n/1000000)}$ lies outside P and is feasible far past any input size of interest. The convention is a chart, and a useful one, but a chart is not the structure it maps. *Structural; the chart named.*

Layer two, the asymptotic tail. The separation is Π^0_2: for all machines, for all exponents, there exists an input on which the machine fails. The collapse is Σ^0_2. Neither direction is finitely witnessable. This is one quantifier register deeper than the Riemann hypothesis, which is Π^0_1 under the Lagarias reformulation and whose *falsity* is therefore witnessed by one bad zero at a finite stage. Nothing about P versus NP is witnessed at a finite stage in either direction. *Type T on the classifications.*

Layer three, the limit object. P is an infinite union over exponents and, by the time hierarchy theorem, equals none of its stages. The union is what manufactures the second quantifier. There is no single machine, and no single exponent, whose behavior the question is about. *Type T on the hierarchy leg.*

Layer four, the worst-case shell. The divergence of Section 3.1. *Conditional-grade, with the missing bridge of Section 3.2 theorem-grade conditional.*

Layer five, oracle non-invariance. Baker, Gill, and Solovay showed in 1975 that the decorated family splits both ways: there is an oracle relative to which the classes separate and one relative to which they coincide. The comparison with the Riemann hypothesis is sharp here. Riemann's analogous decorated family, the function-field zeta functions, was settled by Weil and Deligne and affirms the hypothesis **unanimously**. P versus NP's family splits. *Type T on both sides.*

Over all five layers sits a fact due to Levin. There is an explicit, already-written algorithm, optimal up to a constant factor for NP search problems with checkable witnesses and extended to decision by self-reducibility, and it runs in polynomial time exactly when the collapse holds. The object is not hypothetical and not awaiting construction. It was constructed in 1973 and it is sitting in the literature.

What survives the stripping is therefore a tail-statement about an object already in hand. We will refer to it as **the stripped string**, and the question it poses can be stated in one line: *does the tail of Levin's machine bend, at the one window whose neighbours disagree?* Section 3.5 explains the window and the neighbours.

3.5 · The Window Test: Is the Resource Register Gauge or Structure?

A chart is a description we impose; a structure is what is there whatever we impose. The two are distinguished by an invariance test: vary the chart with the structure held fixed and see whether the reading changes. A reading that varies with the chart alone carries no information about the structure and cannot enter a verdict as a fact about it.

This test is what dissolved layer one above: the feasibility reading varies with the convention while the structure stands, so it is chart. The question is whether the same test dissolves layer two, the resource register itself. Apply it directly. The collapse shape, an existential quantifier over witnesses against deterministic evaluation, has been read at several resource registers. Ask whether the answer is constant across them.

Table 2 | The collapse question read across resource registers. The answer is not constant, so the register is structure and not chart.

Resource register	Answer	Authority
Computability	**Separates**	Halting theorem
Multitape linear time	**Separates**	PPST 1983
Polynomial space	**Collapses**	Savitch 1970
Log space: L vs NL	**Open**	second window
Log space: NL vs coNL	**Collapses**	Immerman-Szelepcsényi
Polynomial time	**Open**	the window

One row above needs its register named, because two different questions live at log space and running them together would manufacture the symmetry this section is testing for. *Collapse* asks deterministic evaluation against existential quantification: that is L versus NL, and it is open. *Complementation* asks whether the nondeterministic class is closed under complement: that is NL versus coNL, and Immerman and Szelepcsényi settled it. Savitch collapses the collapse question at polynomial space; Immerman-Szelepcsényi settles the complementation question at log space. They are not the same result and the table now says so. The correction costs the section nothing and buys it honesty: the collapse question is decided at three registers, separating at two and collapsing at one, with two windows open at log space and at polynomial time.

The answers are not constant. The register therefore carries real information about the answer, and the window is **load-bearing** rather than gauge: the mutual information between the register and the answer is not zero, and no dissolution can strip the window without erasing information the neighbours prove it carries.

This is the decisive disanalogy with the Riemann hypothesis. Riemann's outermost interpretive layer, the critical strip, dissolves cleanly: the strip's width is an output of the chart-selection function and carries zero mutual information with the structure. Once removed, the residue is a decided shape under a blind reader. P versus NP's outermost layers strip identically, but the innermost joint does not: it holds, because the neighbours split around it. One shell over a decided shape; five shells over an undecided one whose innermost joint is real.

3.6 · The Tempering: A Fixed Point One Resource Over

One entry in Table 2 cuts against the reading this paper will develop, and we record it before rather than after we develop it.

Section 4 will show that complementation, the symmetry the collapse question most naturally suggests, is fixed-point-free on languages. A natural inference from that fact runs: a fixed point of complementation would be a class closed under complement, this runs against the grain of the terrain, therefore NP ≠ coNP, therefore the separation. That inference is not available, and Immerman-Szelepcsényi is why. Nondeterministic log space *is* closed under complementation. It is a proven fixed point of the complementation involution, sitting one resource register over from the open window.

So the grain argument fails as stated. Whatever the fixed-point-freeness of complementation on *languages* establishes, it does not establish that a complement-closed *class* cannot exist, because one demonstrably does. The lean survives only as a lean at the time register, and it is permanently weakened by this record. Section 4.4 states exactly what the fixed-point structure does and does not license, and the constraint recorded here is the reason it is stated so narrowly.

4 · The Fixed-Point Structure of the Stripped String

Every question of this kind admits symmetries, and where one of them is **native** in the sense §4.3 makes exact, its fixed-point set is what a large family of proof strategies takes as its target. This section states that criterion, applies it to the Riemann hypothesis and to the stripped string, and finds that the two differ in the one place that matters: one of them has a native symmetry and the other has none. All exhibits are elementary and reproduce from the code printed in Appendix F.

4.1 · What a Native Involution Is, and Why Its Fixed Set Matters

An involution on a content space is a map σ with $\sigma^2 = \mathrm{id}$. Its fixed set $\mathrm{Fix}(\sigma)$ is the subspace of objects equal to their own image. Read linearly on the centered content space, σ splits it into a $+1$ eigenspace, the fixed set, and a -1 eigenspace, the part the involution reverses.

The reason this matters is not decorative. A large and historically dominant family of strategies for settling a conjecture proceeds by *localization followed by residence*: identify the determinate set the symmetry hands you, then prove the objects of interest do or do not sit on it. The strategy needs a target set. If $\mathrm{Fix}(\sigma)$ is nonempty the target exists and the strategy has somewhere to point. If the conjecture names no symmetry's fixed set, the strategy has no target, and this is a fact about the conjecture's form rather than about any particular attempt. Section 4.4 is where this is held to its exact scope, because the weaker-sounding claim that a *terrain* can host no fixed locus at all is false and this paper retires it there.

4.2 · The Riemann Case: A Fixed Set of Dimension One, and What It Costs

Nothing in this section is contested, and a reader who does not carry the Riemann hypothesis in working memory needs only the following.

The object. The Riemann zeta function $\zeta(s)$ takes a complex number $s = \sigma + it$ and returns a complex number, and it encodes the distribution of the prime numbers. It has zeros. The trivial ones sit at the negative even integers and nobody worries about them. The rest, the nontrivial zeros, are known to lie somewhere inside the vertical strip $0 < \sigma < 1$. The Riemann hypothesis is the claim that every one of them lies on the single vertical line $\sigma = 1/2$, the critical line. That is the whole statement, and its shape is the one that matters here: it is a **residence claim**, objects against a line.

Where the line comes from, and why it takes two symmetries rather than one. The line is not chosen and not guessed. It is handed over by symmetry. But it takes two, and an earlier form of this paper said it took one. The correction is worth making in public because it sharpens the argument rather than softening it.

The first symmetry is the functional equation, which relates $\zeta(s)$ to $\zeta(1 - s)$. The associated map is $s \mapsto 1 - s$, an involution since applying it twice returns s. Its fixed set is the solution of $s = 1 - s$, which is $s = 1/2$ and nothing else: **a single point, not a line.** Read as a linear map on the plane centered at 1/2 it is $z \mapsto -z$, minus the identity, eigenvalues −1 and −1, involution residual 0.0 exactly, **fixed dimension zero**. The functional equation on its own therefore carries the *identical signature to complementation* and hands the hypothesis nothing at all.

The second symmetry is Schwarz reflection. ζ is real on the real axis, so $\zeta(\bar{s}) = \mathrm{conj}(\zeta(s))$, and the map $s \mapsto \bar{s}$ carries zeros to zeros. Neither symmetry mentions the zeros. Each is a property of ζ itself.

Compose them. The map $s \mapsto 1 - \bar{s}$ sends (σ, t) to $(1 - \sigma, t)$. It is an involution, since $1 - \mathrm{conj}(1 - \bar{s}) = s$. It carries zeros to zeros, being a composite of two maps that do. And its fixed set is $\sigma = 1 - \sigma$ with t left free, which is **exactly the critical line**. Read linearly on the plane centered at 1/2 it is diag(−1, +1): eigenvalues −1 and +1, residual 0.0 exactly, **fixed dimension one**. The arithmetic is checkable in a line. The first nontrivial zero, $\rho = 1/2 + 14.134725141734693i$, returns $|\rho - (1 - \bar{\rho})| = 0$ exactly, while under the functional equation alone it returns 28.27. The zeros are fixed by the composite and are not fixed by either factor.

So the critical line is **bought**, and the price is a second symmetry. This is the sharper statement, and the argument needs it: Riemann's target set is not free, it is purchased with a reality property of ζ that has no counterpart on the language side.

The structural consequence stands and now rests on the right map. The composite hands the hypothesis its target set **without ever mentioning the zeros**: the functional equation is a property of ζ, Schwarz reflection is a property of ζ, and the line falls out of the two together. The critical line is a determinate, nonempty, one-dimensional object that exists whether or not anyone proves anything about it. The entire open content of the hypothesis is residence: whether the zeros sit on a line that indisputably exists. This paper takes no position on that question; the present author's separate verdict on the Riemann string is recorded elsewhere[45] and is load-bearing on nothing here.

4.3 · The Complexity Case: No Native Involution At All

The criterion first, because §4.2's repair forces it into the open and the earlier form of this section left it unstated.

The criterion. **An involution is native to a conjecture when the conjecture is a residence claim on its fixed set.** That is what native has to mean for the localization-then-residence family to have a target, and §4.2 is the exhibit: the Riemann hypothesis *is* the sentence "the zeros reside on Fix($s \mapsto 1 - \bar{s}$)". Nothing weaker will do. A symmetry that merely acts on the same space is not native. It has to be the one whose fixed set the conjecture names.

Complementation on the language space, $L \mapsto \bar{L}$, is an involution, since double complementation is the identity. It has no fixed points, and the argument is one line: if $L = \bar{L}$ then for any string x we have $x \in L$ iff $x \notin L$, a contradiction. **No language equals its own complement.**

The object-level exhibit makes this concrete rather than abstract. Over the four-string universe {00, 01, 10, 11} there are exactly $2^4 = 16$ languages. Complementation pairs them into exactly **8 two-cycles** with **0 fixed points**, and the involution law comp(comp(L)) = L holds for all 16, checked rather than asserted. Read linearly on the centered indicator space, complementation is minus the identity. Its eigenvalues are {−1, −1, −1, −1}. The involution residual is 0.0 exactly. **The fixed subspace has dimension zero.**

Complementation is native to a conjecture, and it is not this one. *Is NP closed under complement?* is precisely the residence claim NP ∈ Fix(comp) read at the level of classes, and it is a real open question with a real target set: the complement-closed classes, of which P, PSPACE, and by Immerman-Szelepcsényi nondeterministic log space are members. P versus NP is not that question. P = NP entails NP = coNP and the converse is not known, so the two are distinct, and complementation is native to the second. This paper's earlier form ran the two together, and the same conflation is corrected at Table 2 for the same reason.

So what set *is* P versus NP a residence claim on? Theorem 1 answers it and the answer is the whole disanalogy. Over an NP-complete language the conjecture is $L \notin$ P: a residence claim on **P**. And P is not the fixed set of any symmetry specifiable without it. One can of course define an involution that fixes exactly P, by fixing every language in P and pairing off the rest, but that map is P wearing a costume; it is specified *using* the set it is supposed to hand over, and it hands over nothing that was not already assumed.

That is the difference, and it is not a fact about eigenvalues. Riemann's target set is delivered by two symmetries, neither of which mentions the zeros. P versus NP's target set is P, which is the object the conjecture asks about. There is nothing to localize onto, because the thing you would localize onto is the question.

Table 3 | The symmetries of the two questions, computed. The deciding rows are the last two, and neither is an eigenvalue.

Property	Riemann hypothesis	P versus NP
Candidate symmetry	$s \mapsto 1 - \bar{s}$, functional equation ∘ Schwarz reflection	complementation on languages, $L \mapsto \bar{L}$
Specified without mentioning the target objects?	yes, both factors are properties of ζ	yes
Fixed set	$\mathrm{Re}(s) = 1/2$, the critical line	empty
Eigenvalues, centered	-1, $+\mathbf{1}$	$-1, -1, -1, -1$
Involution residual	0.0 exactly	0.0 exactly
Fixed-subspace dimension	**1**	**0**
Object-level exhibit	ρ fixed by the composite at residual 0; 28.27 under the functional equation alone	16 languages, 8 two-cycles, 0 fixed points
Bare functional equation / bare comp	$s \mapsto 1 - s$: fixes the **point** 1/2, dim **0**	dim **0**
Composite with a further symmetry	dim **1**, and it is the target	σ' = comp ∘ flip: dim > 0, and it is **not** the target
What the conjecture resides on	**Fix of the symmetry**	**P, the object in question**
Native involution exists?	**yes**	**none known, and not comp**

The deciding rows are the last two, and neither is an eigenvalue. Both questions admit symmetries; both admit composites with fixed sets. What Riemann has and P versus NP lacks is a symmetry **whose fixed set the conjecture names**. Riemann's line is handed over by two properties of ζ that never mention a zero. P versus NP's target set is P, and you cannot localize onto the thing you are asking about.

4.4 · The Counter-Exhibit, Run Against Ourselves

The strongest objection to §4.3 lands here, and we run it ourselves rather than wait for it, because it defeats a claim this paper made and the claim deserves to be retired in public.

The objection. Section 4.2 grants the Riemann side a *composite* of two symmetries. Grant the language side the same freedom and complementation's fixed-point-freeness evaporates.

It does. Here is the witness, on this paper's own four-string universe. Let *flip* be the map that flips the last bit of every string, an automorphism of the alphabet. Set $\sigma' = \text{comp} \circ \textit{flip}$. It is an involution, since *flip* commutes with comp and each is an involution, and this is checked in Appendix F rather than asserted. And it has **four fixed points**: {00, 10}, {01, 10}, {00, 11}, {01, 11}.

So fixed-point-freeness is a property of the map, not a property of the terrain. Any claim of the form *the terrain hosts no fixed locus for a residence argument to target* is false at the generality the Riemann side is granted, and this paper made that claim. **It is retired here.** No verdict of this paper rests on it, and Section 9.5 records why the composite is not compressible to that claim in the first place.

What survives is the criterion, and it survives cleanly. σ' has a fixed set, and P versus NP is not a residence claim on it: nothing in the conjecture mentions {00, 10} or its analogues at any alphabet size. σ' is therefore not native either, by §4.3's criterion, and neither is complementation. The fixed sets a composite manufactures are not targets, because no conjecture points at them. Riemann's composite differs in exactly the respect that carries the argument: **its fixed set is the object the hypothesis names.** The test is not *does a fixed set exist*, which anyone can arrange, but *does the conjecture reside on it*, which no arrangement supplies.

The eigenstructure is therefore an **exhibit and not the reason**, and it is a good exhibit. Complementation is the one symmetry structurally tied to the collapse question, since swapping a language for its complement is exactly what turns an existential quantifier over witnesses into a universal one. That symmetry, the natural one, the one the question itself suggests, hands over nothing: fixed set empty, dimension zero, sixteen languages in eight two-cycles. The reason P versus NP has no target is the conjecture's form, per §4.3. The reason its most natural symmetry cannot rescue the situation is the eigenstructure. The first is load-bearing. The second makes it concrete.

What is established, and it is theorem-grade because it is trivial: complementation on languages is fixed-point-free, the associated linear map is $-I$, and Fix has dimension zero.

What follows structurally: the localization-then-residence family of strategies has no target here, because the set it would aim at is P and P is the object in question. That is a fact about the conjecture's form, holds now and held before anyone tried, and covers strategies of that family not yet invented, since the absence of an independently specifiable target is not the kind of thing a new technique supplies.

What does not follow, and the record is explicit: the fixed-point-freeness of complementation on *languages* says nothing about complement-closure of *classes*. Nondeterministic log space is closed under complementation, by Immerman and Szelepcsényi. It is a proven fixed point of the induced map on classes, sitting one resource register away from the open window. Any argument of the form *a complement-closed class runs against the grain of the terrain, therefore NP ≠ coNP, therefore separation* is refuted by that theorem, and we do not make it. Nothing here forecloses a proof of the separation by a strategy that is not a residence argument. Section 5 assembles the record of what has been shown about the strategies that are not.

4.5 · A Second Type-Indexed Fact: The Squared Invariant Is Mute on Direction

One further structural fact belongs here because it constrains the present paper's own instrument before it constrains anyone else's.

The verification operator of Section 6 returns a verdict from the determinant of a correlation matrix built from the three warrant rows, and that determinant equals the square of a signed scalar. Reflecting any single axis conjugates the correlation matrix by a diagonal sign matrix D, and $\det(DRD) = \det(D)^2 \cdot \det(R) = \det(R)$. The determinant is therefore **bit-identical under negation of the claim**, while the signed scalar it squares flips sign.

Executed at seed 20260622 over 24 contexts: the baseline returns $\lambda = -0.965027450462$ and $\det(R) = 0.931277980144$; the reflected frame returns $\lambda = +0.965027450462$ and $\det(R) = 0.931277980144$. The difference $|\det(R)\_P - \det(R)\_\neg P|$ is 0.0 exactly. The λ ratio is -1.000000 exactly. The kernel identity $|\lambda^2 - \det(R)|$ closes at 5.551×10^{-16}, with $\kappa(R) = 1.6568$, far inside the conditioning gate.

The consequence binds this paper's instrument and every instrument of its class: **a rotation-invariant functional of the warrant geometry cannot carry the truth-sign.** It certifies the dimensionality of the support, never its direction. The catalogue of such functionals is closed, by Weyl, inside the real-part subring of quaternion words, so this is not a limitation of one construction but an identity over all of them.

This is why the operator of Section 6 never emits a verdict on the stripped string. It reads magnitude on rows it did not author, and the direction, where a direction is recoverable at all, is carried by the linguistic decomposition and the physical arrow, not by the number. Section 10.2 returns to this as the paper's own instrument being strapped before any other.

5 · Results I: The Barrier Landscape, Fifteen Results in Three Classes

The literature names three barriers. The record holds more, and the number is less important than the sorting. This section assembles the complete record and organizes it on an axis that is not, to our knowledge, the usual one: **the temporal structure of each barrier relative to the technique it blocks**. That axis is what carries Section 10, and it is the reason the assembly is a result rather than a survey.

5.1 · The Organizing Axis

A barrier is a theorem stating that some class of arguments cannot settle the question. Barriers differ in what they quantify over, and the difference is decisive.

A **technique-indexed** barrier quantifies over a named class of techniques. It says: *no argument of this described form decides the question*. Such a barrier must wait for its technique class to be described before it can be proven, so it necessarily arrives after the class it blocks. It is complete over the techniques that exist and silent over those that do not, and a genuinely novel technique escapes it not by defeating it but by falling outside its quantifier.

A **type-indexed** barrier quantifies over the structure any technique must engage. It says: *whatever the argument is, it must engage this object, and this object has this property*. Such a barrier does not wait for anything. It has zero lag and zero foresight, and it covers techniques not yet invented on exactly the same terms as those already invented, because it never quantified over techniques in the first place.

The distinction is not a matter of strength. A technique-indexed barrier can be unconditional and a type-indexed one can be conditional. It is a matter of what the theorem's quantifier ranges over, and therefore of what a new idea has to do to get past it. Getting past a technique-indexed barrier requires being a new kind of technique. Getting past a type-indexed barrier requires the structure to be other than it is.

5.2 · Class I: The Technique-Indexed Barriers, Six Rows

Every row of this class arrived after the technique family it blocks. The lags are recorded and they are the class's signature.

Row I.1 · Relativization. Baker, Gill, and Solovay proved in 1975 that there exist oracles A and B with $P^{A} = NP^{A}$ and $P^{B} \neq NP^{B}$. An argument relativizes if it stays valid when every machine in its statement and proof is given an arbitrary oracle. Since a relativizing statement holds relative to both A and B, and any statement deciding the question would assert one verdict relative to every oracle uniformly, no relativizing argument decides it. This blocks diagonalization and essentially every classical computability-style separation. *Generation one; lag ten years behind the diagonalization era. Type T, unconditional.*

Row I.2 · Natural proofs. Razborov and Rudich proved in 1994 that no proof strategy that is both constructive and large can establish the circuit lower bounds required, if pseudorandom function families of exponential security exist. This blocks essentially the whole combinatorial lower-bound program. *Generation two; lag nine years behind the circuit era. Type C, conditional on the cryptographic hypothesis.* This row is also the class's most interesting member, and Section 5.3 explains why: it is self-referential in content, since the truth of strong hardness is precisely what eats a class of its own proofs.

Row I.3 · Algebrization. Aaronson and Wigderson proved in 2008 that arithmetization with algebraic oracle access does not decide the question. This blocks the toolkit behind the celebrated interactive-proof and probabilistically-checkable-proof results, which is to say the toolkit that had just finished producing the field's most striking non-relativizing successes. *Generation three; lag eighteen years behind the arithmetization triumph. Type T.*

Row I.4 · Locality. The locality barrier, in the 2020 conference form and the 2022 journal form, shows that local reductions and local magnification arguments cannot by themselves cross. It arrived two years behind hardness magnification, the shortest lag on record and by a wide margin. *Generation five. Type T within its stated scope.*

Row I.5 · The cousin row, three wings. Geometric complexity theory was constructed explicitly to evade Rows I.1 through I.3, and it has drawn its own walls. *Generation four. Wing a, occurrence obstructions:* Bürgisser, Ikenmeyer, and Panova proved in the 2016 conference form, journal form 2019, that occurrence obstructions cannot separate the determinant-permanent pair, fifteen years behind the program's founding. *Type T. Wing b, rank methods:* Efremenko, Garg, Oliveira, and Wigderson established in 2018 that rank-based lower-bound methods hit stated ceilings. *Type T within scope. Wing c, algebraic natural proofs:* Forbes, Shpilka, and Volk, and independently Grochow, Kumar, Saks, and Saraf, showed in the 2017 conference form, journal form 2018, that the natural-proofs shape is reborn in the algebraic register. *Type C.*

Row I.6 · The proof-systems track. Razborov proved in 1995 that superpolynomial circuit lower bounds are unprovable in named fragments of bounded arithmetic under strong pseudorandomness assumptions. This row is read on the ladder itself rather than on the technique, and it is the only Class I row of that shape. *Type C.*

Table 4 | Class I, the six technique-indexed barriers, enumerated on eight lines since Row I.5 carries three wings. Every row is a chaser.

Row	Barrier	Authority	Lag	Warrant
I.1	Relativization	Baker-Gill-Solovay 1975	10 yr	Type T, unconditional
I.2	Natural proofs	Razborov-Rudich 1994; JCSS 1997	9 yr	Type C, on the crypto hypothesis
I.3	Algebrization	Aaronson-Wigderson 2008	18 yr	Type T
I.4	Locality	CHOPRS 2020; JACM 2022	2 yr	Type T within scope
I.5a	Occurrence obstructions	Bürgisser-Ikenmeyer-Panova 2016; JAMS 2019	15 yr	Type T
I.5b	Rank methods	Efremenko-Garg-Oliveira-Wigderson 2018	not dated	Type T within scope
I.5c	Algebraic natural proofs	Forbes-Shpilka-Volk; GKSS 2017; Theory of Computing 2018	not dated	Type C
I.6	Bounded-arithmetic unprovability	Razborov 1995	ladder-read	Type C

The lag record across the five dated generations reads **ten, nine, eighteen, fifteen, two years**. This is the class's temporal law and Section 10.1 is where it does its work.

5.3 · The Generator Laws of Class I

Four laws govern this class, and each is stated at its honest grade because two of them are frequently overstated.

The pair-birth law. A formalized method class is an invariance signature: it is defined by what it does not distinguish. An invariance signature is a blindness. The blindness is the wall. On this reading classes and walls are born paired, and the wall's arrival is not luck but the consequence of the class having been formalized enough to be described. *Corroboration-grade on the five-generation record. The universal law is unproven and we state it as unproven.* What the record shows is five for five; what it does not show is that the sixth is guaranteed.

The reflexivity amplifier. The stripped string quantifies over the machine space, and the machine space contains the provers, the proof-search procedures, and Levin's enumerator. Every formalized efficient method is therefore inside the problem's own quantifier domain from birth. Gödel's 1956 letter to von Neumann is the birth certificate of exactly this reading: the original question was literally the running time of a machine searching for proofs. Row I.2 is the proven instance of the amplifier, since strong hardness is what eats a class of its own proofs. This is the sharpest structural contrast with the Riemann hypothesis, whose string quantifies over zeros and whose domain therefore excludes its readers. *Structural, with Row I.2 the theorem-grade instance.*

The orbit-mixing mechanism and its terminus. Each wall arises because the method class mixes orbits the answer distinguishes. The regress of walls cannot become total by this mechanism, because total coverage of every method class would be independence, and independence is a theorem nobody has. *Structural.* This law is the reason Class I cannot, by its own machinery, ever close.

The retired generator. One tempting generator is refuted and we record the refutation rather than the temptation. Transcribed from the recorded arc battery over a contingency table this paper does not print, the mutual information between *presupposes the foundation* and *walled* is **0 bits**, while the mutual information between *carries an invariance signature* and *walled* is **1 bit**. Appendix F states the constraints those two figures carry and why the law below does not rest on them. Foundational presence beneath everything triggers nothing in particular; the walls are invariance-indexed. No appeal to foundational presence ever explains, predicts, or constitutes a barrier, and this retirement is permanent. *Engineering-grade on the computation; structural on the law.*

5.4 · Class II: The Type-Indexed Barriers, Eight Rows

Every row of this class has zero lag, because none quantifies over a technique.

Row II.1 · The fixed-point-free involution. Section 4.3. Complementation is fixed-point-free; eigenvalues all -1; residual 0.0; fixed subspace dimension 0. No residence argument has a target, born or unborn, this paper's own instrument first among them. *Type T on the eigenstructure; structural on the reading, scoped per Section 4.4.*

Row II.2 · Orientation-blindness of the squared invariant. Section 4.5. $\det(DRD) = \det(R)$ is an identity, not a policy. Any rotation-invariant functional of the warrant geometry is mute on the truth-sign, and the catalogue of such functionals is closed by Weyl. *Type T.*

Row II.3 · The physical bar. The mutual information between the stripped string and the physical register sits at the independence floor, computed at $\mathbf{7.19 \times 10^{-5}}$, and the thermodynamic arrow forbids pre-actuating a search. No physical process, no thermodynamic argument, and no reading of the arrow settles the string. Sections 7 and 8 prove this row in full. *Type T at the floor and the arrow; engineering-grade at the estimator.*

Row II.4 · The misattribution bar. The 0-bit-against-1-bit computation of Section 5.3, seated here as a standing type-check on every future wall attribution. *Engineering on the computation; structural on the law.*

Row II.5 · The metatheoretic rung law. Every metatheory rung is recursively axiomatized, arithmetic-bearing, and consistent, hence each carries its own unprovable consistency; the regress is productive and terminal at no stage, by Turing 1939 and Feferman 1962. Every derivation-bearing standpoint is therefore barred rung by rung. The one stratum that is not barred is not a theory, generates no derivations, and is occupied by no prover, so it contributes no proof. *Type T on the per-rung application; structural on the stratum placement.*

Row II.6 · The access structure. The access structure is placement-forced: no deciding input is generated from the verdict side, and the exception is typed exclusively as *supplied* input. The wall-and-aperture machine executes it: the state is closed under all eight verdict-side actions, which are exertion, extrapolation, preference, any method born or unborn, the metatheory tower, evidence accumulation, this architecture itself, and the direction-recovering instruments; and the state changes under exactly one input class, supply. A wall and an aperture are indistinguishable under every verdict-side action and are distinguished by exactly one thing, which side's action changes the state. *Operational-procedural on the parse; engineering on the machine.*

Row II.7 · The jurisdiction type-check. No resolution of P versus NP grounds, proves, or contains the foundation the analysis rests on, since the set of formal proofs of a foundation from a base weaker than itself is provably empty. The row blocks one lane in one direction and leaves the problem lane untouched, which is what makes it a type-check rather than a barrier on the problem. *Type T on the type-check.*

Row II.8 · The soundness cell, and the citability law. Consider the cell *sighted and walled*: a route that has been seen to resolve the target, and against which a wall has been asserted. That cell is **empty**, because a wall against a route that resolves the target would assert a falsehood, and a sound theory proves no falsehoods. The row's entire legitimate content is a bar on one closing argument: **Class I is never citable as the total wall.** It certifies no route, earns no route's existence, and lives here only as a barrier on an argument. *Type T on the cell; structural on the placement.* This row governs every citation of Class I in perpetuity, including every citation in this paper.

5.4.1 · "Absolute" Is Typed, Not Rhetorical

The word carries an exact taxonomy and we give it before we use it. A row of this class is **type-based and timeless** where the barrier is a type-check, and it covers the unborn identically with the born because it never quantified over techniques. It is **constitutive** where the barrier is an identity, and no ingenuity makes an identity false. It is **theorem-eternal within scope** where the barrier is a named theorem, for exactly as long as that theorem stands, which is permanently. Rows II.1, II.2, II.3 at the floor and the arrow, II.5, and II.8 are Type T; II.3's estimator, II.4's computation, and II.6's machine are engineering-grade; II.6's parse is operational-procedural. Every use of the word below is one of these three types and none of them is a mood.

5.4.2 · The Assembled Statement

The eight rows are not eight observations. They assemble, and the assembly is the result.

From every standpoint the verdict side occupies, **access to the resolution of P versus NP is zero**. The object ladder is barred by its own diagonal. The metatheory tower is barred rung by rung, each rung recursively axiomatized and arithmetic-bearing and therefore carrying its own unprovable consistency. This paper's own instrument is barred twice over, by the absent target of Row II.1 and by its own orientation-blindness at Row II.2. Physics is barred at the floor and the arrow, at a mutual information of exactly zero by §9.3, the recorded estimator sitting at its own noise floor of 7.19×10^{-5}. Evidence is barred by the register law that no convergence becomes proof. Every named method family is barred within scope for as long as its theorem stands. And every mouth, this one first, is barred from totality in either direction by Row III.1.

The deciding input is constitutively supplied and never generated. Coverage over the enumerated standpoints is one hundred percent, current-complete and indexed as such by Row III.1, this paper's own kernel strapped first. The answer is determinate on the $\mathbb{N}$-truth premise at its exact cap, and it is held where no verdict-side standpoint is.

This is consistent with the absence of an independence theorem, and the consistency is not a hedge but a difference of quantifier. Coverage-on-the-verdict-side is a claim about *access from standpoints*: no standpoint the analysis occupies reaches the resolution by its own action, and Rows II.1 through II.8 are the enumeration of the standpoints and the bar on each. It is **not** the claim that no proof exists, which would be independence, which is a theorem nobody has. The two quantify over different sets, and the bridge between them is the typing of the input: a proof, if one comes, arrives as a **supplied deed** and not as a reading executed from where the analysis stands. That is why Row II.6's machine finds the state closed under all eight verdict-side actions and changed by exactly one input class. **A wall and an aperture are indistinguishable under every verdict-side action, and are distinguished by exactly one thing: which side's action changes the state.**

5.5 · Class III: The Closure Row

Row III.1 · The closure barrier. Consider the operator *certify-completely*, which takes a barrier catalogue and returns the assertion that the catalogue is finished. This operator has **no fixed point**, and the reason is not length but structure. The certifying instrument is born with a blindspot, by the pair-birth law of Section 5.3. The list it certifies complete therefore cannot contain the row it cannot itself see. Every total certificate spawns exactly one row it cannot contain, and the chain of certified catalogues is strictly increasing: **10, 11, 12, 13, 14, 15** under five iterations, never stabilizing. The increment is one because the row is one, per this row's own statement; an earlier form of this paper printed an increment of two and attributed the loop to a seed, and Appendix F's Exhibit D refutes both, since the loop consumes no randomness and carries no seed at all.

Total self-inclusion, the container placed inside its own contents, is the map $x \mapsto -x$ on the content space. Its eigenvalues are all -1, its involution residual is 0.0 exactly, and its fixed subspace has dimension **zero**. This is the identical signature computed in Section 4.3 for the problem's own native involution. **One engine, verified twice, at the bottom of the stack and at the top.** The same absence of a fixed locus that forbids a language from equalling its complement below forbids the catalogue from containing its own head above.

Four theorem anchors wear one image: Gödel's second incompleteness theorem, the system-level instance; Tarski's undefinability theorem, the language-level instance; the ungroundability result, that a posited foundation is not promotable to a theorem of its own base; and the machine-zero eigenstructure just computed.

The anatomy is exact and both halves matter. **The tail is swallowable**: a catalogue may contain rows about its own earlier rows, and a partial self-catalogue is consistent, exhibited here by direct construction. **The head never is**: the total certificate does not exist. The exit is not through the mouth but upward, and it is a growth law rather than a door. The strictly increasing chain is the Turing-Feferman tower read on the barrier census: catalogue $n+1$ certifies catalogue n and arrives carrying its own new blindspot. Turing built the consistency tower deliberately in 1939; Feferman proved in 1962 that the transfinite progression is complete for the Π^0_1 truths, with the completeness bought entirely by the path through the ordinal notations and owned at no single stage. The regress is productive and is not a Münchhausen trap: each rung is real progress, each stage census is a real result, and the absolute certificate is the one thing the tower never delivers, because delivering it is exactly what the diagonal forbids. **The catalogue is completable relative to any stage and finalizable at none.** *Type T on the fixed-point-freeness; structural on the placement.*

Row III.1 binds this paper before it binds anything else. No claim in this paper asserts that the barrier record is finished, and Section 5.6 states the completeness verdict in the only form this row permits.

5.6 · The Completeness Verdict

Over the barriers that exist: current-complete. Every wall with a theorem behind it is seated in a row above. The three-barrier account standard in the literature undercounts the field by a factor of five.

Over the barriers that do not yet exist: no verdict issues, by Row III.1. The only pronouncement this assembly makes about its own future census is none. The catalogue's file changes state under exactly the same supply-typed inputs as the question it guards, and no verdict-side event closes it.

5.7 · The Unwalled Region

The complement of the walled region is surveyed as an intersection, and we survey it rather than describe it. A method surviving every Class I row lies outside the relativizing class, outside the natural-proofs shape in both the Boolean and the algebraic registers, outside the algebrizing class, outside the local class, outside occurrence obstructions, and outside the stated rank ceilings. The region is not empty: interactive-proof results are non-relativizing, and geometric complexity theory was built to be non-natural. **No occupant of the**

full intersection is certified, and this paper says nothing further about the region beyond its coordinates on the map. Certifying an occupant is precisely what Row II.8 forbids anyone to do from this side, and it is what Row III.1 forbids the catalogue to claim about itself.

6 · The Verification Operator and Its Two Ranks

The audit apparatus has an exact statement as a three-valued decision rule on the sign of a determinant. We state it at the strength it earns, which is less than it is usually credited with, and we state the one place where it hands off rather than deciding.

6.1 · The Encoding, the Criterion, and the Margin

An encoding assigns to a proposition a real $3 \times N$ matrix whose rows are the three warrant streams over N reading contexts. Rows are centered and scaled to unit variance; admissible covariates are removed by orthogonal projection; the Gram matrix G is the residualized encoding times its transpose over $N - 1$.

The load-bearing fact is standard. $\det(G)$ is the squared volume of the parallelepiped the rows span, and it is strictly positive exactly when the rows are linearly independent. By Cauchy-Binet it equals the sum of squared 3×3 minors over column triples, divided by $(N - 1)^3$ under the normalization used here, so it is nonnegative and vanishes exactly on degeneracy.

The closed form binds the geometry to an algebra. Normalize the residual rows to unit length, read each as a pure quaternion, and set $\lambda = \mathrm{Re}(\hat{q}_1\hat{q}_2\hat{q}_3)$. Then $\mathbf{det}(\boldsymbol{R}) = \boldsymbol{\lambda}^2$, where R is the correlation matrix, and $\mathbf{det}(\boldsymbol{G}) = \boldsymbol{d_1 d_2 d_3} \cdot \mathbf{det}(\boldsymbol{R})$ with d_a the residual variances. The first identity holds because the scalar part of a product of three pure quaternions is the negative scalar triple product of their imaginary parts, and the Gram determinant of three unit vectors is the squared triple product. The verdict is bounded, $0 \le \det(R) \le 1$ by Hadamard, and it is invariant under rotation of the axes, since the quaternion automorphisms are inner and the real part is conjugation-invariant. The axis labels are therefore gauge.

The margin is exact. The smallest singular value of the encoding is the distance to the nearest rank-two matrix, by Eckart-Young; each Gram eigenvalue moves by at most the spectral norm of any symmetric perturbation, by Weyl. The verdict that the smallest eigenvalue is positive therefore persists under every perturbation below it. **The margin of a seal is its distance to degeneracy, and nothing else about the entries is material.**

6.2 · What the Determinant Certifies, and What It Does Not

A strictly positive determinant certifies **linear non-degeneracy of the chosen encoding**: no row is a linear combination of the others. This is strictly weaker than functional non-recoverability, and the gap is exhibited rather than asserted. In $\mathbb{R}^5$ take $x = (1,2,3,4,5)$ and $y = (1,4,9,16,25)$. Every entry of y is a deterministic function of the corresponding entry of x, so the pair is not functionally non-recoverable. Yet the 2×2 Gram determinant is $55 \cdot 979 - 225^2 = 3220 \neq 0$, so the vectors are linearly independent. **A nonlinear deterministic dependence is invisible to the determinant.**

Functional non-recoverability is therefore a modelling assumption asserted by the deletion discipline, never a consequence proven by the determinant. Wherever this paper's encoding is asserted to satisfy the criterion by appeal to non-recoverability, the claim is a hypothesis on the encoding.

One earlier statement of this operator described the sealed configuration as three orthogonal axes summing to zero by cancellation. That object does not exist, and the correction is what gives the criterion a real referent: three mutually orthogonal nonzero vectors **cannot** sum to zero, since by the Pythagorean identity the squared norm of the sum is the sum of the squared norms and is strictly positive. A vanishing sum of nonzero vectors is by definition a nontrivial linear dependence, which forces the determinant to zero rather than above it. The genuine cancellation lives one level down, in the residualization, where the projection plus the residual minus the original vanishes identically with three individually nonzero and non-orthogonal summands. **The cancellation is a sum that vanishes; the seal is a volume that does not.**

6.3 · The Two Ranks, and Why They Are Independent

Two quantities the operator certifies must not be conflated, and their independence is the paper's cleanest structural instrument.

The **encoding rank** is the rank of the residualized encoding, equivalently the number of strictly positive Gram eigenvalues. It is a relation *among the rows*: do they span a full three-volume? It is exactly what the determinant sees, and encoding rank three is equivalent to $\det(R) > 0$, equivalently $\lambda \neq 0$.

The **discrimination rank against a target** is the number of rows carrying strictly positive mutual information with that target in the favouring direction. It is a relation *between the rows and an external indicator*: does the row change with the truth of the target? It is exactly what the evidence criterion of Section 2.3 counts.

These are logically independent, and the independence is realized rather than hypothesized. Against the stripped string, Section 9.3 shows the physical and registrational rows are invariant: their conditional-mean shift is zero and their likelihood ratio is one. Yet the same three rows remain linearly independent as vectors, since linear independence is unaffected by whether a row co-varies with an external indicator. **Encoding rank three with discrimination rank one is not a contradiction. It is the generic situation for this target.**

The verdict economy reads both ranks jointly, and it stays three-valued while doing so: the readings below are not four peer verdicts but the three verdicts with their targets typed, and the consistency result in particular is a **seal whose target is the conditional** rather than the proposition, which is the same move §9.3 makes for Face 3 and is stated here so the economy is not read as growing a fourth cell.

A full seal on a target requires encoding rank three *and* discrimination rank three. A consistency result holds when the encoding rank is three and the discrimination rank is below three with the deficit closed by an explicitly admitted premise; it certifies consistency under that premise and non-degeneracy of the encoding, and nothing further. A collapse is encoding rank below three, independently of any target. And a proposition whose only favouring row is logically equivalent to the target itself is routed out of the operator entirely, since the operator declines to adjudicate a claim on warrant equivalent to the claim.

6.4 · Where the Operator Hands Off

The operator does not emit the terminal verdict of Section 9, and the reason is Section 4.5: the determinant it reads is orientation-blind by an identity, so it cannot carry a truth-sign. It certifies the dimensionality of support, never its direction.

The routing is therefore explicit. Where the operator routes the stripped string out of band as conclusion-dependent, adjudication passes to the five-gate admission protocol of Appendix C, which is a separate emitter reading the terrain rather than the warrant geometry. The operator and the protocol never overlap: the operator reads rows, the protocol reads the fixed-point structure, the barrier record, and the aperture. **The number is a receipt, not a verdict.** This division is what Section 4.5 called the paper's own instrument being strapped first.

7 · Results II: The Bridge Premise Is the Conjecture

The empirical record establishes that no bounded physical system performs exhaustive search at scale. The question is what single inference would carry that fact to the class separation. This section identifies the inference exactly and proves it is the conjecture.

Definition. The **no-shortcut premise** C is the proposition that the NP-complete language admits no deciding algorithm running in polynomial time by any method.

The premise is precisely what must be added, and the point of the next theorem is that it is not a step toward the conjecture but the conjecture itself, so we state the gap without pretending the premise is smaller than it is. The physical fact bears on exhaustive search alone and is silent on whether a polynomial algorithm exists. Any bridge must therefore **supply** the missing proposition outright rather than derive it, and the proposition it must supply is that the language admits no polynomial-time decider by any method. **The premise is the entire logical distance, and the physical conjunct contributes nothing to the crossing.** An earlier form of this paragraph glossed the premise as *no method other than essentially exhaustive search decides the language*, which is a different and weaker sentence; the gloss is retired, since the two are equivalent only under a reading on which the theorem below is immediate, and the theorem is the point rather than an inconvenience.

Theorem 1. *Over an NP-complete language L, the no-shortcut premise is logically equivalent to P ≠ NP.*

Proof. By Cook-Levin, if L is NP-complete then $L \in \mathrm{P}$ iff P = NP: forward, membership in P together with polynomial-time reducibility of every NP language to L places every NP language in P, and the reverse inclusion holds always; backward, if the classes coincide then L, being in NP, is in P. The premise C is by definition the proposition $L \notin \mathrm{P}$. Negating the equivalence, $\neg C$ is equivalent to $L \in \mathrm{P}$, which is equivalent to P = NP. Therefore C is equivalent to P ≠ NP. ∎

The consequence is a measurement rather than a prohibition. The distance between a real physical fact and the abstract separation is **one premise wide, and that premise is identically the conjecture**. The equivalence is not approximate, not up to a reduction, and not modulo a hypothesis. Any argument proposing to cross from the physical asymmetry to the separation must supply the premise, and supplying it is supplying P ≠ NP. An argument that crosses while claiming not to have assumed the conjecture has either smuggled the premise under another name or committed a non sequitur at the gap.

8 · Results III: The Bridge Has No Physical Discharge

Theorem 2 (the universal bar). *No physical argument discharges the no-shortcut premise unless it proves the conjecture.*

Proof. To discharge the premise is to establish that the language admits no polynomial-time decider by any method, which by Theorem 1 is establishing P ≠ NP. A physical argument that discharged it would therefore be a proof of the conjecture. ∎

This covers every physical route, whatever resource it invokes, and it is the general reason. A second and sharper bar covers the specific bounds at issue.

Proposition 3 (the relativization bar). *Counting-type physical bounds cannot discharge the premise.*

Proof. A lower bound derived solely from a count of physical operations or of distinguishable physical states counts elementary steps and counts simultaneously distinguishable subsystems; it does not inspect what any step computes. Granting each step oracle access leaves both counts unchanged, since an oracle query is one elementary step producing one bit, and the distinguishable-state count is a function of energy and radius alone. Such a bound therefore relativizes. By Row I.1 there is an oracle relative to which the classes coincide and one relative to which they separate; a relativizing statement holds relative to both; a statement deciding the conjecture would assert one verdict relative to every oracle uniformly. Contradiction. ∎

The scope of Proposition 3 is stated precisely, since overclaiming it would be the error this paper exists to expose. **Relativization covers counting-type arguments and no others.** A physical argument exploiting non-relativizing structure, a spectral or algebraic or instance-specific property rather than a raw count, is not blocked by Row I.1, and we do not claim it is. An adiabatic argument resting on a spectral-gap property of a problem-specific Hamiltonian is a non-counting physical argument and lies outside the relativization bar. Its permanence is supplied instead by Theorem 2: a spectral argument that discharged the premise would, like any other, prove the conjecture.

The division of labour is deliberate. **Theorem 2 is the universal bar and covers every physical route. Proposition 3 is the sharper bar and covers the counting routes specifically.** The premise standing between the physical asymmetry and the separation has, demonstrably, no physical foundation from which it could be supplied, and the localization is stable under every future physical advance, because the gap it names is the conjecture and the conjecture is not a physical quantity.

9 · The Composite Verdict

The audit returns three verdicts on three faces of the proposition, plus one fence standing beside them. The faces do not merge, and Section 9.4 explains why the tuple is not compressible to a single token.

9.1 · Face One · The Physical Field · Sealed

The generate-verify asymmetry is a completed physical event read where it is real. Checking is cheap and finding is dear across the actualized computational field: search is charged by the nonequilibrium work relations, bounded in rate by the quantum speed limits, and bounded in parallel extent by the distinguishable-state count, while the thermodynamic arrow forbids pre-actuating a search whose result is not yet available. This is sealed on the event and its arrow at the physical stream, and on no supplied string.

The face is **answer-independent**, and this is the point. It stands identically in the world where the classes separate and in the world where they coincide but no polynomial algorithm has been found. The rows are literally constant functions of the truth of the separation, by the invariance argument of §9.3, so their mutual information with the stripped string is **exactly zero**, proved rather than estimated. A finite-sample plug-in estimator does not return exact zero on independent variables; it returns its own positive bias floor, and the recorded arc battery returns 7.19×10^{-5} at its declared seed and sample size. That number is the estimator's noise floor and it **confirms the proof rather than competing with it**; it is engineering-grade, it is not reproduced in this paper, and no claim here rests on it. *Sealed; theorem-grade at the floor and the arrow; scope-fenced from the string by the invariance.*

The fence on this face is load-bearing and we state it before anyone else has to. The floor bounds the cost of each mandatory event; it does not bound the *number* of mandatory events. Reversible computation severs per-step cost from operation count, so the floor cannot be summed into an exponential bound on a computation that may be arranged reversibly. And a count of configurations is not a lower bound on the steps required to decide a property of them: the determinant of an $n \times n$ matrix is a sum of $n!$ signed terms and is computed in cubic arithmetic by elimination. **Face one bars physical brute-force search. It does not reach the string.**

9.2 · Face Two · The Geometric Shape · Sealed

The problem is the stratal collapse question itself: deterministic evaluation against existential quantification over witnesses, the inclusions matching the strata. Its most natural symmetry is complementation, native to a different question by §4.3's criterion, and Section 4.3 computed it: fixed-point-free, eigenvalues all -1 exactly, residual 0.0 exactly, **fixed-subspace dimension zero**. No symmetry delivers this conjecture a target, by structure and not by ignorance: the target is P, the object in question.

The reflexive membership is timeless: every formalized efficient method is inside the problem's own quantifier domain from birth, and Gödel's 1956 letter is the birth certificate. This is a **shape fact, decided**, and it is silent on which way the string falls. *Sealed at theorem grade on the eigenstructure.*

9.3 · Face Three · The Formal String · The Terminal Verdict

The stripped string of Section 3.4 is determinate on $\mathbb{N}$-truth at a stated premise cap, and the openness of that string is located in the terrain rather than in the search. Section 10 states what this means in full, and it is the paper's central claim. Here we record the two computations that establish the face and the one that fixes its type.

The truth-invariance computation. Let H be the indicator of the separation, one in the world where the classes separate and zero in the world where they coincide but no polynomial algorithm has been exhibited. Let the physical row be a deterministic function of the execution traces of the algorithms actually run, and the registrational row a function of a verifier's check-cheap generate-costly architecture. Both take identical values under both values of H, hence carry **zero mutual information** with the truth of the separation. The physical row is determined by the execution trace of each fixed algorithm on each input, which is fixed by the algorithm and input alone and not by whether a faster algorithm exists unexecuted elsewhere; holding the execution history fixed, its distribution is invariant under H, the conditional entropy collapses to the marginal, and the mutual information is zero. The registrational row is a function of a fixed architectural property and the identical argument applies. The conditioning on a fixed execution horizon is necessary and we state it: unconditionally, a world in which the classes coincide would over time make discovery of a fast algorithm more likely, so the invariance is asserted at a horizon at which no polynomial decider has been run.

The rank computation. Against the stripped string the discrimination rank is therefore **one**: only the structural row shifts with H, and the row that shifts is the admitted premise, which by Theorem 1 is the conjecture. The encoding rank remains **three**, since linear independence among rows is unaffected by whether a row co-varies with an external indicator. Encoding rank three with discrimination rank one is the generic situation for this target and it is why no full seal is available on it.

The type, and the seal. The verdict on this face is not the ordinary open verdict, and Section 10 establishes the difference rather than asserting it. In the ordinary open verdict the file is incomplete and the repair is named. Here **the file is complete**: the object is stripped, the terrain is measured at machine zero, the barrier record is assembled in three classes, the aperture is located, and what all of that shows is that the openness is a property of the terrain.

The verdict, stated exactly, because the token is precise and the ordinary word for its cell is not. The economy of §2.2 admits three verdicts and no fourth, and this paper holds itself to it. [$Ø_0$] is a **refinement inside the open cell**, not a fourth verdict. But §2.2 also says what that cell's refinements are, and it says it in the sentence that matters: the ordinary open verdict is where *the file is incomplete and the repair is named*, and a refinement is where *the file is complete and the reader nonetheless cannot read the answer*. Those are different findings. Reporting the second by printing the first's name would be the report this paper exists to refuse.

The verdict on the string is [$Ø_0$], the grounded-sealed halt. It is not ordinary openness. Ordinary openness says: nobody has found it yet, keep looking, and here is the repair. [$Ø_0$] says the opposite of each clause. The file is complete: the string is determinate on ℕ-truth at the stated premise cap, and no independence proof exists, so this is not a Ghost either, the term for a proposition proved undecidable in both directions and therefore carrying no determinate value to reach. The obstruction is proven rather than surveyed: the conjecture's target set is P, the object in question, per §4.3, which is a fact about the conjecture's form and not a report about effort. The formulation space is exhausted rather than sampled: the fifteen-row record sorts by quantifier, per §5, and the type-indexed class binds the unborn on the terms it binds the born. The companions are decided: two faces seal beside this one. And the one door is named rather than left implicit: the aperture of §10.3, a determinacy witness supplied from outside, which this paper locates and does not cross.

What is sealed is the halt, and the distinction carries the paper. A proof of either direction is not offered here and is not claimed, and the rank computation above is why it cannot be: encoding rank three with discrimination rank one is not a seal, and §6.3 says so in the same words. So the string's truth-value is not delivered. What *is* delivered, and sealed, is that **no standpoint that exists reaches it**, and that the reason is structural rather than evidential. It is **terminal by structure, not by fatigue**. Nothing about the halt is pending, nothing is provisional, and no part of it awaits a further diagnostic. By Row III.1 it makes, and licenses, no claim of permanence in either direction, and Section 10.4 is where that is made exact.

The halt-proposition carries its own rank pair, and it is not the string's. §6.3 states the criterion and this paper owes it on every target it seals, this one included. Against the target *no standpoint that exists reaches the string*, run §2.3's test row by row. The structural row shifts: §4.3's form and §5's fifteen-row record bear on reachability and not on which way the string falls. The physical row does **not** shift, and it fails here for the same reason it fails on the string: in a world where a proof is unreachable, observed solver scaling is S, and in a world where a proof is reachable but unfound, observed solver scaling is identically S, so nothing physical co-varies with the reachability of a proof and the likelihood ratio is one, the same two-world invariance §9.3 proves for the string one paragraph up. An earlier form of this paragraph counted the physical **bar** of Row II.3 as a shift in the physical **row**; the bar's discriminating content is Theorem 2 and Proposition 3, which are theorems, and §2.1's isolation test reassigns theorem-vocabulary content to the structural stream, so counting it physically double-counts one piece of evidence across two streams, which is the exact error the isolation test exists to prevent. The registrational row shifts by Row II.6's machine, whose content, which side's action changes the state and the object already in hand, is registrational by §2.1's definition, and it is argued rather than computed as Appendix F records. **Encoding rank three, discrimination rank two.** By §6.3 the halt therefore takes the **consistency result: a seal on the conditional**, its deficit closed by the explicitly admitted ℕ-truth determinacy premise, which is the cap Table 1 already prints on this face and which every structural conclusion in this paper inherits. This is not a retreat, and the arithmetic is the reason: a rank of three here would be the claim that the physical face reaches the halt, which §§7 through 9.1 spend three sections proving it does not, and the seal on the conditional at the declared premise is the rank-pair image of exactly the cap the halt's own admission carries. The target-indexing of §2.3 is what allows one instrument to return (3,1) against the string and (3,2) against the halt without contradiction, and the two pairs differing is the whole reason the split is available rather than a convenience.

Why the inversion does not need more than this. §1.2's inversion needs the *court* closed, which the sealed halt is. It never needed the string proved, and a paper claiming that would be claiming a proof it does not have. The court being closed is what leaves the structural determination holding the only verdict available at any standpoint that exists.

9.4 · Beside the Faces · The Fence

One verdict stands beside the faces and never among them. It is **broken**, and it is scoped to named inflations only, terminal for none of the faces.

Broken: the claim that a physical collapse of the classes is a mechanism. Broken: the claim that the fixed-point structure forces the separation as a formal proof, which Section 4.4 retires by the Immerman-Szelepcsényi record. Broken: any claim of permanent immunity, and equally any claim that the barrier record is finally complete. The last pair is Row III.1 executing, and it devours both signs symmetrically.

The proposition that the classes coincide takes the ordinary broken verdict on the evidence, and its **type is load-bearing**: it is broken for **absence of positive warrant on every axis**, and not for demonstrated falsity. No construction exists, no measurement favours it, no registrational witness supports it, and in every restricted **circuit** model where the question is formally resolvable it resolves toward separation, the qualifier being load-bearing since Table 2 records that the space registers do not. But an analysis that does not establish the separation cannot, on pain of contradiction, establish its negation false. The asymmetry between the two propositions is an asymmetry of **available warrant under identical handling**, not of established truth value.

9.5 · The Composite, and Why It Does Not Compress

Table 5 | The composite. Three faces on the proposition, one fence beside. The tuple does not compress.

Object	Structural	Physical	Registrational	Verdict	Grade
Face 1 · the physical field	n/a	work relations, speed limits, the arrow	witnessed generate-verify asymmetry	**Sealed**, scope-fenced	theorem, fenced
Face 2 · the geometric shape	fixed-point-free involution, Fix dim 0	n/a	n/a	**Sealed**	theorem on the eigenstructure
Face 3 · the formal string	discrimination rank 1, the premise = the conjecture	truth-invariant, MI **exactly 0** by §9.3	truth-invariant, likelihood ratio 1	**[$\varnothing_0$] Sealed halt**, in the terrain	structural, on Type-T legs / premise cap
Beside · the named inflations	n/a	n/a	n/a	**Broken**	scoped to the inflations
Beside · the coincidence claim	no favouring stream	likelihood ratio 1	likelihood ratio 1	**Broken**	absence of warrant, not falsity
The bridge premise $\Longleftrightarrow$ separation	Cook-Levin identity	n/a	n/a	**Sealed**	theorem, Section 7
A physical discharge of the bridge	none	counting bounds relativize	n/a	**Broken**	theorem, Section 8

The tuple is **not compressible to a single token**, and the reason is Row III.1. Collapsing three faces into one final verdict would be a totalizing pronouncement about the whole, and the mouth issuing it would be claiming to contain its own head. Any citation of this paper that collapses the faces misquotes it.

One reading of Table 5 closes the question the rows would otherwise leave open. The column asymmetry, the separation carrying two sealed faces while the coincidence claim carries nothing positive, is an asymmetry of warrant **across distinct targets**, not two rival answers to one question with opposite established truth values. By the truth-invariance computation the physical face and the coincidence claim are **co-satisfiable at the working horizon**: a world in which the classes secretly coincide but no polynomial algorithm has been found still exhibits every physical fact face one asserts. The faces therefore seal on warrant the coincidence claim cannot muster, the coincidence claim breaks for want of warrant, and these are jointly consistent precisely because they name different targets.

10 · Results IV: What a Terminal Verdict Means, and Why It Is Not a Prediction

This is the paper's central claim and it needs the most care, because it is the place where an overstatement would be indistinguishable from the confidence it is trying to earn.

10.1 · Two Species of Openness

An open question can be open in two structurally different ways.

Openness of the first kind is epistemic. The diagnostic is incomplete. Something has not been found, and what has not been found might be found. The Goldbach conjecture and the twin-prime conjecture rest here: there is no theorem stating that the available instruments cannot decide them, and their openness is a record of what has been tried. The honest verdict is the ordinary open one, and it names the repair: keep working.

Openness of the second kind is structural. The diagnostic is *complete*. The object has been stripped to its arithmetic residue, the terrain has been measured, the barrier record has been assembled and sorted, and the aperture has been located. And what the completed diagnostic shows is that the openness is a property of the terrain rather than a deficiency of the search.

The difference is not a difference of confidence or of degree. It is a difference in what the analysis has *done*. In the first case the file is incomplete. In the second the file is complete and the reader still cannot read the answer, and the analysis says exactly why in terms that are checkable.

P versus NP's stripped string sits in the second case, and the whole of Sections 3 through 8 is the demonstration. This is also where the sibling comparison earns its keep. The Riemann hypothesis's string is also open of the second kind, and yet the two are different species. Riemann's hypothesis resides on a line a symmetry delivers, dimension one: the reader is blind before a line that indisputably exists. P versus NP resides on P, the object in question, and no symmetry delivers a line at all, dimension zero: there is no line to be blind before. **Blindness in the reader; walls in the terrain.** That one measured bit, computed in Section 4, is the whole taxonomy, and §2.2 is where the economy requires it to be: the two refinements inside the open cell differ by exactly one measured quantity, and this is that quantity.

Section 4.4's demotion does not reach this and the distinction is worth one sentence, because the two claims are easy to run together and they are not the same claim. The eigenstructure is **not the reason** P versus NP has no target; that is the conjecture's form, per §4.3, and it is what survives the counter-exhibit. The eigenstructure **is the discriminant** that sorts which species of halt the string sits in, and here it is load-bearing exactly as §2.2 requires. A symmetry that delivers a line and a symmetry that delivers nothing produce different species of openness, and one bit tells them apart.

10.2 · The Criterion: Which Barriers a New Technique Defeats

The objection that must be met is obvious and correct in shape: *every barrier so far has been circumvented by a technique nobody anticipated, so why is this different?* The objection is a good one and the answer is not confidence. It is a criterion.

Consider what it takes to get past a barrier of each class.

A technique-indexed barrier is defeated by being a new kind of technique. Relativization says no argument *of a described form* decides the question. The interactive-proof results were not blocked by it and did not defeat it; they fell outside its quantifier by being non-relativizing. Natural proofs says no argument that is constructive and large works, under a hypothesis. Geometric complexity theory was constructed precisely to fall outside that quantifier. In both cases the barrier stands, unrefuted, and the technique goes around it, and this is exactly what a technique-indexed barrier permits: it never quantified over techniques it had not seen. **A new technique escapes it by existing.**

A type-indexed barrier is not defeated by being a new technique, because it never quantified over techniques. The fixed-point-freeness of complementation does not say *no argument of this form works*. It says: any residence argument requires a determinate target set, and there is no such set. A new technique does not supply a target set. It would have to make the structure other than it is, and the structure is that no language equals its own complement, which is a one-line consequence of what complementation is. Orientation-blindness does not say *this functional is weak*. It says $\det(DRD) = \det(R)$, an identity over all constructions whatever, with the catalogue of such functionals closed by Weyl. There is no ingenuity that makes an identity false.

The criterion is therefore exact and it is not a prediction: **ask what the barrier's quantifier ranges over.** If it ranges over descriptions of methods, a new description escapes it, and the record shows this happening on a reliable schedule. If it ranges over the structure any method must engage, then a new method engages the same structure, and the barrier binds it on the identical terms it binds every method already tried. The second kind has **zero lag and zero foresight**: it does not wait for a technique to be described, it binds at utterance, and it bound the question before anyone asked it, because a structural fact has no publication date.

This is what makes the diagnosis future-proof in the only sense the word can honestly carry. It is not a forecast that nobody will succeed. It is the observation that the type-indexed rows are not the kind of thing succeeding gets past, because they are not claims about attempts.

10.3 · What the Terminal Verdict Licenses

Four clauses, and each is a consequence of Section 10.2 rather than an attitude.

Method-terminal. The verdict licenses no iteration of failed methods. This is not the counsel of despair; it is the difference between a wall that is *felt* and a wall that is *typed*. A history of failed attempts licenses trying harder. A theorem about the method class licenses trying something else. Every Class I row is of the second kind and names precisely which family it retires.

Reframe-terminal. No restatement moves the verdict. Verdicts move on new structural argument or new mathematical mass, never on rephrasing, and this discipline binds symmetrically: the field's conviction that the separation holds carries no mass, and any author's conviction that the terrain is closed carries none either.

Instrument-terminal. No new instrument of the same type crosses, by the criterion of Section 10.2. This clause is scoped exactly to the type-indexed rows and to nothing else. It does not say no instrument crosses; it says the type-indexed rows do not distinguish between instruments born and unborn, so a new instrument gains nothing merely by being new.

Terminal-for-the-record. This is the most important clause and the most misread. If the question is one day resolved, the resolution **replaces** this verdict and does not **correct** it. The verdict was, and remains, the correct reading of the terrain as mapped at issuance. It never claimed the terrain could not change under a supplied input; it claimed the terrain as measured has these properties, and it does. A crossing supplies new terrain rather than revealing that the reading was wrong. This is what distinguishes a structural verdict from a bet: a bet is falsified by the outcome, a reading of a mapped terrain is superseded by new terrain.

10.4 · What the Terminal Verdict Does Not License

The clause that keeps the previous four honest is stronger than all of them, and it binds this paper first.

It is not independence. No independence theorem exists for this problem, and we do not claim one. Total coverage of every method class *would* be independence, and independence is a theorem nobody has. Row II.8 forbids citing the Class I record as the total wall, and that row governs every citation in this paper including this sentence.

It is not a permanence claim. By Row III.1 the operator *certify-completely* has no fixed point. The assertion *this barrier record is finished* is not available from inside the record, now or later. Every total certificate spawns a row it cannot contain, and the chain is strictly increasing at 10, 11, 12, 13, 14, 15 without stabilizing. The exit is upward and it is a growth law: the catalogue is completable relative to any stage and finalizable at none.

It is not fate-terminal, in either sign. Row III.1 devours *permanently immune* and *finally complete* with the identical mechanism and without term. The pessimist's totalizing sentence and the optimist's are the same sentence with the sign flipped, and the same fixed-point-freeness eats both. This paper makes neither, and the strap falls on this paper's own mouth before it falls on anyone else's.

It is not broken geometry on the problem. The time hierarchy stands; the terrain hosts separations; at least one complement-closed class exists in it. Nothing here says the question has no answer. The answer is determinate on $\mathbb{N}$-truth at a stated premise cap, and the cap is exactly the cap the Riemann string carries. That is the one axis on which the two questions are equal.

10.5 · The One Sentence About the Door

The aperture is located, typed as supplied rather than generated, and uncrossed.

That sentence is required and it is load-bearing: the admission protocol of Appendix C was executed in both directions, and with the sentence present the verdict is admitted, while with the sentence deleted Gate 4 fails and the verdict demotes to the ordinary openness. The absolute statement of Section 5.4.2 therefore stands at full strength **because of** that sentence and not despite it. This paper says nothing further of the door. Every imagined occupant of its far side is an afterimage and is fenced as such, contemplated nowhere here. **The silence on the far side is a strap and not a hedge.**

11 · The Sibling Comparison: Two Millennium Strings on One Skeleton

The Riemann hypothesis is the natural control for every claim in this paper, because it is open, famous, arithmetic, and carries a native involution, and it differs from P versus NP in exactly the places the analysis says matter. Every row below is drawn from a result established above or cited to its source.

Table 6 | The two strings compared. The rows the analysis says matter are the ones that differ.

Property	Riemann hypothesis	P versus NP
Symmetry that delivers the target set	$s \mapsto 1 - \bar{s}$, functional equation ∘ Schwarz	**none known**
Fixed set of that symmetry	the critical line, dim **1**	comp: empty, dim **0**
What the conjecture resides on	**the fixed set**	**P, the object in question**
Clean arithmetic form	Π^0_1 (Lagarias)	Π^0_2 separation, Σ^0_2 collapse
Finite witnessability	falsity witnessed by one bad zero	neither direction witnessed
Determinacy at the Ground	determinate on $\mathbb{N}$-truth, premise-capped	determinate, **the identical cap**
Object already in hand	every zero computable one by one	Levin's machine, explicit since 1973
Domain against readers	the zeros **exclude** the provers	the machines **include** the provers
Interpretive shells	one, the critical strip	five
Innermost joint	**pure gauge**: strip width dissolves under remap	**load-bearing**: the neighbours split around it
Sibling family	function fields: **unanimously affirm** (Weil, Deligne)	oracle decorations: **split both ways** (BGS)
Self-limiting twists	none of the reflexive kind	Razborov-Rudich; Levin's object-in-hand
Undecidability theorem	none exists	none exists
Species of openness	blindness in the reader	walls in the terrain

Four of these rows deserve a sentence, because they are the ones that turn the comparison into an argument rather than a table.

The strip against the window. Riemann's outermost shell dissolves under the invariance test of Section 3.5: the critical strip's width is an output of the chart and carries zero information about the structure. P versus NP's outermost shells dissolve identically, but the innermost joint does not, because Table 2 shows the answer is not constant across resource registers. One shell over a decided shape; five shells over an undecided one whose innermost joint holds.

The family split. Riemann's decorated family was settled and it affirms. P versus NP's decorated family was settled and it splits both ways. The same instrument, pointed at the two problems, returns unanimity in one case and contradiction in the other.

The domain. Riemann's string quantifies over zeros, and provers are not zeros. P versus NP's string quantifies over machines, and provers *are* machines. The string swallows its own readers, and Razborov-Rudich is that fact proven at one instance.

The one axis of equality. Both strings are determinate on $\mathbb{N}$-truth at the identical premise cap, and both answers are held where no verdict-side standpoint is. The entire difference between the two arcs is the conjecture's form: **one resides on a set a symmetry delivers, and the other resides on the question itself.** The one measured bit of §2.2, the dimension of the delivered set, one against zero, is that fact made concrete and it is what sorts the two species of halt.

12 · Discussion, and the Limitations That Bound the Claim

On the barrier record. The literature's three-barrier account undercounts by a factor of five, and the undercount matters because it hides the sorting. Three barriers read as three obstacles. Fifteen results in three classes read as a landscape with a temporal law, and the law is what Section 10.2 needs.

On the prior programs. Deolalikar in 2010 and Blum in 2017 both claimed the separation via circuit lower bounds and both were retracted after community review found fatal errors. Both were single-stream attempts inside the natural-proofs quantifier. Geometric complexity theory remains the most technically sophisticated live program and it does not claim resolution; it has drawn Row I.5a against itself, which on the pair-birth reading is what a program does when it becomes formalized enough to be described.

On what this paper settles. Three seals and the results carrying them, each at the grade its section earns, none provisional, and not one of them requiring a line of invented mathematics.

Settled, Type T: the bridge premise between any physical resource bound and the class separation **is** the conjecture, by Cook-Levin, exactly and not up to a reduction (Section 7). *Settled, Type T:* that premise has **no physical discharge**, universally by Theorem 2 and, for counting-type bounds specifically, by relativization at Proposition 3 (Section 8). *Settled, Type T:* the symmetry the stripped string most naturally suggests, complementation, is **fixed-point-free**, eigenvalues all minus one, residual 0.0 exactly, fixed subspace of dimension **zero**, against the Riemann involution's dimension one, exhibited over the four-string universe at sixteen languages, eight two-cycles, zero fixed points (Section 4). *Settled, Type T:* the squared invariant is **mute on the truth-sign** by an identity over all constructions, catalogue closed by Weyl (Section 4.5). *Settled, and the assembly is the result:* **access to the resolution from every standpoint the verdict side occupies is zero**, the deciding input constitutively supplied and never generated, coverage over the enumerated standpoints one hundred percent, current-complete and indexed as such by Row III.1 with this paper's own instrument strapped first (Section 5.4.2). *Settled, structural on a complete diagnostic:* the openness of the string is **located in the terrain**, and the verdict on it is sealed, terminal by structure rather than by fatigue, against method, against reframing, and against instrument (Sections 9.3, 10).

Settled, and it is the paper's reason for existing: **the inversion.** Structural evidence is subordinate to proof only while a proof is reachable. Row 3 establishes that none is, from any standpoint the verdict side occupies, terminally. The subordination clause therefore has no referent here, and the structural determination stands as the only determination available anywhere a verdict can issue from. Its grade stays non-deductive and it never becomes a theorem; what changes is that it is no longer **provisional**, because there is nothing left for it to be provisional to.

Two of the paper's three faces seal outright and the third takes [$\varnothing_0$], the grounded-sealed halt, which is §2.2's terminal refinement inside the open cell and is the same cell Table 1 names: the file is complete, the halt is sealed, and a proof of the string is not claimed. The barrier record is **current-complete over the barriers that exist**, and the literature's three-barrier account undercounts it by a factor of five. None of this is pending. None of it is provisional. There is no wait-and-see anywhere in it.

On what this paper does not settle. No proof of either direction is offered. No independence result is claimed and none exists. The Clay criteria are not met. Section 4.4 records that the fixed-point structure does not force the separation, and Section 3.6 records the theorem that refutes the natural argument that it does. Section 9.4 records that the coincidence claim breaks for absence of warrant and not for falsity. Section 10.4 records that the terminal verdict is neither independence nor permanence, and that Row III.1 forbids the record from claiming its own completeness.

On the conditional grades. Four load-bearing legs are conditional and we name them. The natural-proofs row is conditional on the cryptographic hypothesis. The Heuristica witness is conditional on the five-worlds partition being genuine. The Bogdanov-Trevisan wall is conditional and bars only the non-adaptive lane. The determinacy of the string on $\mathbb{N}$-truth is premise-grade, and every structural conclusion inherits that cap.

On the instrument. The verification operator certifies linear non-degeneracy of an encoding, which is strictly weaker than functional non-recoverability, and the counterexample is exhibited in Section 6.2. The encoding is a modelling choice and is not canonical. The operator's determinant is orientation-blind by an identity, which is why the operator never emits the verdict of Section 9.3 and hands off instead.

On the one asymmetry a critic should press. The physical face counts three streams; the string counts one. Section 9.3 gives the computation, and the honest exposure is that the structural row's likelihood ratio against the *physical* face is argued rather than computed: restricted lower bounds are formal theorems whose truth-value does not vary across physical worlds, and their favouring of the physical face rests on the claim that they raise the probability that observed scaling reflects structural hardness rather than weak solvers. That claim is defensible and it is the softest leg in the paper. If it fails, the physical face drops to discrimination rank two and takes the seal on the conditional rather than the seal on the proposition, per §6.3's typing. Nothing else in the paper moves.

13 · Falsifiable Predictions

The physical face is a forecast with consequences, and it is falsifiable across three instrument families with no shared calibration lineage.

Prediction 1 · Substrate-agnostic gradient. The work and execution time of actual solvers navigating an NP-complete configuration space scale exponentially in problem size while verification scales polynomially, and the gradient between the two is invariant across substrate classes. *Method:* micro-calorimetry and transition-time measurement on physical solvers, against verification on identical hardware, calibrated against colloidal-trap measurements of single-operation cost. *Null:* a substrate on which the observed generation-verification gradient vanishes or inverts. An earlier form of this prediction forecast a **minimal-work** lower bound and named sub-exponential aggregate dissipation as a falsifier. That form is retired and the retirement is forced by this paper's own fence: §9.1 and Appendix A both record that aggregate dissipation is severable from operation count by reversible arrangement, so Bennett's construction realizes the null without touching the face, and the predicted bound was the summed floor the paper elsewhere retires. The prediction is restated on the anchors Appendix A keeps, the observed gradient, which is what Appendix B's physical stream actually carries.

Prediction 2 · Biological network separation. Generative search and verification of a pre-computed certificate occupy distinct functional configurations that mutually inhibit. *Method:* combined fMRI and MEG on generative problem-solving against certificate checking. *Null:* simultaneous activation and seamless functional overlap.

Prediction 3 · The estimator floor. The mutual information estimated between any physical observable of a running solver and the truth of the separation remains at the estimator's own bias floor under every experimental refinement, falling toward zero as the sample grows rather than rising. *Method:* direct estimation on instrumented solver runs across substrate classes, with the floor calibrated on synthetic independent pairs at matched sample size. *Null:* a measured mutual information rising with instrument precision and sample size. The typing is exact and it is narrower than the previous form of this prediction claimed: this **audits the encoding, not the world**. §9.3 proves the true value is zero, so the prediction cannot test that; what it tests is whether the estimator on real instrumentation behaves as it does on known-independent inputs. A rising value would not refute the proof; it would show the physical row is not the constant function §9.3 takes it to be, which is a defect in the encoding and is where the refutation would land.

The three span calorimetry, neuroimaging, and information estimation. Confirmation of all three converts the physical face from theorem-plus-induction to directly measured across substrate classes. It does not touch the string, by Section 9.1's own fence.

14 · Conclusion

The P versus NP question has been recorded as open for fifty-five years, and *open* is an epistemic report about what has not been found. This paper returns two seals on two faces, a sealed diagnosis of why the third stays open, and no proof of either direction, establishes them in four checkable steps, and invents no mathematics anywhere in the process.

The famous question and the arithmetic string are two objects, and the identification between them is unsupported in every world, contradicted in an unrefuted one, and blocked along its natural repair route. Five interpretive layers strip away and what survives is a Π^0_2 tail-statement about a machine constructed in 1973.

The Riemann hypothesis resides on a set that two symmetries hand over. The critical line is the fixed set of $s \mapsto 1 - \bar{s}$, the functional equation composed with Schwarz reflection, and neither factor mentions a zero; the functional equation alone fixes only the point 1/2 and carries the identical signature to complementation, so Riemann's line is bought and not free. P versus NP has no symmetry of that kind. By Theorem 1 its target set is P, the object under question. Its most natural symmetry, complementation, is fixed-point-free: eigenvalues all minus one, residual zero, fixed subspace of dimension zero, exhibited over the four-string universe as sixteen languages in eight two-cycles. Grant the language side a composite and fixed points appear, which is why the eigenstructure is the exhibit and not the reason: the composite's fixed set is not what the conjecture names either. **One question resides on a set a symmetry delivers. The other resides on the question.** That is the difference, and it is not an eigenvalue.

The barrier record holds fifteen results in three classes. Six are technique-indexed and arrived at lags of ten, nine, eighteen, fifteen, and two years behind the families they block; they are complete over what exists and silent over what does not. Eight are type-indexed: they quantify over the structure any technique must engage rather than over techniques, so they have zero lag and bind the unborn on the terms they bind the born. One is a closure result, and it shows the catalogue cannot certify its own completeness, from which it follows that no claim of permanence is available here to anyone, this paper first.

The single premise separating any physical resource bound from the class separation is logically identical to the conjecture, and it has no physical discharge. The gap between a physical fact and the abstract question is one premise wide, and the premise is the conjecture.

The verdict is three faces and a fence. The physical field is sealed and answer-independent, its mutual information with the string exactly zero by §9.3, the recorded estimator sitting at its own noise floor of 7.19×10^{-5}. The geometric shape is sealed and silent on which way the string falls. The string takes [$\varnothing_0$], the grounded-sealed halt, and what is sealed is the halt rather than the string, on the conditional at the declared premise cap: the diagnostic is complete, and what it shows is that the openness is a property of what is there rather than a deficiency of what has been done. Terminal by structure and not by fatigue, against method, against reframing, and against instrument. From every standpoint the verdict side occupies, access to the resolution is zero, and the deciding input is constitutively supplied. And by the closure result, terminal against no one's fate, in either sign, this paper's mouth strapped before any other.

And that is the inversion the paper exists for. Structural evidence is provisional only while a proof might arrive to overrule it. Row 3 establishes that from every standpoint the verdict side occupies, none can. So the determination the geometry returns is not the junior partner awaiting a senior ruling; it is the only verdict available anywhere a verdict can be issued from. The layer that is ordinarily supreme has forfeited, by a theorem, and handed the ruling to the layer that is ordinarily provisional, on its way out.

The forfeit was not extracted from mathematics against its will. It was **derived from it**: complementation's fixed-point-freeness is one classical line, the barrier record is fifteen theorems every one of which belongs to somebody else, and the quantifier classification is standard. Mathematics is what proves mathematics cannot rule here. Nothing was invented to make that happen, and that is why the transfer costs nothing and why the arrangement is checkable in an afternoon by anyone who knows the classical results.

The question is not closed. It is **mapped, the map is finished on the side the map is drawn from, and on that side the map is the territory.** Those are different from closure, and the difference is the paper.

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Appendix A · The Physical Bounds, at Their Honest Scope

Four bounds carry the physical face, and each is stated with the scope it actually has rather than the scope it is usually given.

The per-operation cost floor. The nonequilibrium work relations of Jarzynski and Crooks charge any transition that occurs, with Landauer's bound and its colloidal-trap confirmation by Bérut and colleagues the irreversible-erasure subclass. *Scope:* this bounds the cost of each mandatory event and says nothing about how many events are mandatory. Bennett's reversibility result severs per-step cost from operation count, so the floor cannot be summed into an aggregate exponential bound. This is the fence of Section 9.1 and it is load-bearing.

The rate bound. The Mandelstam-Tamm and Margolus-Levitin limits bound the rate of any transition that occurs by the available energy. *Scope:* a rate bound on transitions, not a count of them.

The parallel-extent bound. The Bekenstein bound, as extended by Bousso, caps the number of simultaneously distinguishable states in a bounded region by its energy and radius. *Scope:* this bars a physically instantiated exhaustive parallel search over an exponential space. It does not bar a polynomial algorithm, and by Proposition 3 it relativizes, since the count is a function of energy and radius alone.

The arrow. The second law forbids pre-actuating a search whose result is not yet available: the trajectory that finds is thermodynamically distinct from the trajectory that checks, and the distinction is directed. *Scope:* this is what makes the physical face carry a direction at all, since by Section 4.5 the squared invariant cannot. It is a fact about actualized computational events and it does not reach the string, by the mutual-information floor of Section 9.1.

The migration recorded for the record: an earlier form of this analysis rested the physical face on dissipation. That anchor is wrong and is retired. Dissipation is avoidable by reversible arrangement, so a dissipation-based bound is escapable by construction. The anchor is operation counts and distinguishable-state counts, which are not avoidable by reversibility, and the face is stated on those.

Appendix B · A Worked Instance

The apparatus is exhibited end to end on one instance so the reader sees exactly what it returns and where it stops.

Object. Boolean satisfiability, the canonical NP-complete language, at the physical face.

Intake. The famous formulation is refused: its vocabulary is engineering-practical and denotes no arithmetic sentence. The stripped residue is admitted. *Section 3.3.*

Streams. Structural: NP-completeness by Cook-Levin, plus the restricted lower bounds of Håstad and Smolensky, plus the barrier record of Section 5. Physical: observed exponential scaling of solver work against polynomial verification on identical hardware. Registrational: the statistical zero-knowledge regime, where the simulation guarantee is information-theoretic and rests on no hardness assumption. The last point is not incidental. A registrational stream resting on a computational zero-knowledge guarantee would presuppose the hardness the analysis is auditing, and the circularity would sink the stream. The statistical regime carries the same architectural content with no such presupposition.

Deletion test. Remove the structural stream: the physical scaling and the verifier architecture remain and neither reconstructs NP-completeness. Remove the physical: the theorems and the architecture remain and neither predicts the measured work. Remove the registrational: neither remaining stream reconstructs the simulation guarantee. Three passes.

Isolation test. The three vocabularies do not back-translate: reductions and lower bounds; joules and seconds; simulator transcripts. Pass.

The halt-proposition's encoding, admitted separately. Its registrational row is Row II.6's machine rather than the zero-knowledge architecture above, so §2.1's two tests are run again on it rather than inherited. *Deletion:* remove the registrational stream and five of Row II.6's eight verdict-side actions are recoverable from the structural row, since methods born or unborn are barred by Classes I and II, the metatheory tower by Row II.5, this architecture by Rows II.1 and II.2, the direction-recovering instruments by Row II.2, and evidence accumulation by the register law. **Three are not.** Exertion, extrapolation, and preference are barred by §2.4's mass requirement and by no barrier row, so their closure is not recoverable from the structural record and the row carries independent support: the deletion test passes on a residue of exactly three actions. *Isolation:* that residue is conviction yielding no capacity, which is §2.1's registrational subject matter stated in its own words, and it does not back-translate into theorem vocabulary. Both tests pass, and the (3, 2) of §9.3 is **admitted rather than assumed**.

Ranks. Against the physical face: encoding rank three, discrimination rank three, **sealed**. Against the halt-proposition: encoding rank three, discrimination rank two, the physical row silent by the two-world test, taking the seal on the conditional at the premise cap per §9.3. Against the stripped string: encoding rank three, discrimination rank one by the computation of Section 9.3, **routed out of the operator** and handed to the protocol of Appendix C.

Where it stops. The instance seals the physical face and returns nothing on the string. That is the whole apparatus working correctly, and an instance that returned a verdict on the string would be the apparatus failing.

Appendix C · The Five-Gate Admission Protocol for the Terminal Verdict

The terminal verdict of Section 9.3 is emitted by a separate protocol, not by the operator, and the protocol's bar is the highest in the economy because the verdict asserts a measured structural fact about the terrain on top of a determinacy claim and a theorem record. Five gates, conjunctive. Defaults absent throughout, so an under-specified call never emits the verdict. Four riders travel as disciplines and are never counted as gates.

The count is forced, not chosen. The terminal node has exactly five neighbours in the verdict graph: the sibling terminal verdict, the dissolve-openness, the ordinary openness, the resolved states, and the broken state. Five neighbours, five boundaries, one gate per boundary. There is no sixth gate, and the count is not made to mirror the sibling protocol's, which answers a different admission question and carries a different forcing. Inflating one count to match the other would be a fitted number, which is the error this protocol is built to avoid.

Gate 1 · The form, and the router bit. Apply §4.3's criterion: is the conjecture a residence claim on the fixed set of a symmetry specifiable **without** its target set? The gate routes on the answer and the four branches are exhaustive.

Branch one, a native symmetry exists with fixed dimension one or more. Compute the dimension at machine zero with an object-level exhibit. This routes the candidate **out of this protocol to the sibling**, because a delivered locus changes the species of the halt from walls in the terrain to blindness in the reader, and neither protocol emits the other's verdict. The Riemann hypothesis exits here: $s \mapsto 1 - \bar{s}$ is native by the criterion, its fixed set is the critical line, dimension one.

Branch one-b, a native symmetry exists with fixed dimension zero. The candidate **continues here**, and the case is not vacuous: a conjecture of the form *the objects reside on* $\mathrm{Fix}(\sigma)$ with $\mathrm{Fix}(\sigma)$ empty is the coherent claim *there are no such objects*, a target that happens to be empty rather than no target at all, and it can be as open as anything, since a nonexistence claim is decided by nothing about the set alone. The branch is distinct from branch two in bookkeeping and not in routing: the localization family has a target and the target is empty, so the family's move degenerates to a nonexistence proof and it has nowhere to point in exactly the way branch two has nowhere to point. Gates 2 through 5 decide the species; dimension alone does not. No candidate in this paper occupies this branch, which is stated because a partition with an unoccupied cell is still a partition and one with a missing cell is not.

Branch two, no native symmetry exists. The localization-then-residence family has no target, and the candidate **continues here**. This is the flagship's branch and §4.3 is the reason: by Theorem 1 the conjecture resides on P, the object in question, and no symmetry specifiable without P delivers it. The symmetry the question most naturally suggests, complementation, is then measured for the record and returns eigenvalues all minus one exactly, involution residual 0.0 exactly, fixed subspace dimension zero, with the sixteen-language exhibit: the natural candidate delivers nothing either, which is the confirming measurement and not the routing reason.

Branch three, nothing measurable. No symmetry identified and no criterion applied routes **ordinary open**: the form is unread and nothing is emitted.

This gate is the switch, and the measured dimension is §2.2's one quantity: one against zero is what sorts blindness in the reader from walls in the terrain. The four branches partition the answers exhaustively: a native symmetry at dimension one or more, a native symmetry at dimension zero, no native symmetry, and no reading at all. *Failure boundary: the sibling verdict.*

Gate 2 · The determinate stripped string. Run the invariance audit first: shell inventory, intake refusal of the famous vocabulary, gauge test on the innermost joint. The residue must be a determinate arithmetic sentence at a stated grade. The joint must be load-bearing, that is, the answer non-constant across the read neighbouring registers. A joint proven pure gauge over an already-decided shape is the sibling's signature and cross-routes. An object that dissolves entirely with no determinate residue routes to the dissolve-openness, the question retyped as chart artifact. *Failure boundary: the dissolve-openness.*

Gate 3 · The wall record, theorem-grade. Blockage must be proven, never felt. At least one theorem-grade wall on a named method class, with citation and scope, or one type-indexed wall applying to the candidate. **A history of failed attempts is not blockage**, and openness without walls routes to the ordinary open verdict, which is where wall-less conjectures rest whatever their fame. Two laws govern the record: walls are invariance-indexed, and the technique-indexed record is never citable as the total wall. *Failure boundary: the ordinary openness.*

Gate 4 · The aperture, crossed or held at one sentence. The witness check runs first: a supplied proof or disproof adjudicates by direction and exits to the resolved states; an instrument-generated witness is rejected at intake. Absent a witness, the aperture is located and typed supply-only, its exits enumerated, and the emission carries exactly one sentence saying so. *Failure boundary: the resolved states, which is not a failure but the exit this gate hands off cleanly.*

Gate 5 · The totality discipline. A supplied independence proof reroutes the candidate to broken, both directions permitted by proof. Absent it, the emission binds under the closure result: terminal for the record only, any crossing replacing and never correcting, and no fate-terminal pronouncement in either sign, with the emitting analysis bound first. *Failure boundary: the broken state.*

The four riders. *R1, face-scoping:* the verdict attaches to one face and never stands alone; at least one sealed companion face is required, and this paper carries two. *R2, machine precision:* every numeric leg executed and float-clean at a declared seed; an escalation-band case emits engineering-incomplete and never the verdict. *R3, revision survival:* the emission holds under reframing pressure in both directions before it issues. *R4, register separation:* nothing outside the structural register travels with the verdict. A confidence assessment, however hedged, is not a structural finding and does not accompany one. This paper issues none, by construction.

The flagship walk. *Gate 1:* branch two, no native symmetry exists, since by Theorem 1 the conjecture resides on P and no symmetry specifiable without P delivers it; complementation measured for the record at eigenvalues all −1 exactly, residual 0.0 exactly, fixed dimension 0, sixteen languages in eight two-cycles, which is the confirming measurement and not the routing reason; **passed**, the router holding the candidate here. *Gate 2:* five shells stripped, the famous vocabulary refused and the Π^0_2 string admitted, determinate on $\mathbb{N}$-truth at the stated cap, the window proven load-bearing by Table 2; **passed**. *Gate 3:* fifteen rows across three classes, the technique-indexed record at lags ten, nine, eighteen, fifteen, two, the type-indexed rows timeless, the misattribution bar at exactly zero bits; **passed**. *Gate 4:* no witness in hand, aperture located and typed supply-only, one sentence emitted; **held**. *Gate 5:* no independence proof exists, the closure result binding the emission, both fate-pronouncements devoured; **passed**. *Riders:* two sealed companion faces; every numeric float-clean at the declared seed; held under sustained reframing across the arc; nothing outside the structural register carried. **Verdict: admitted.**

The negative controls, the gates doing their work. *The Riemann string:* fixed subspace dimension one, the critical line the fixed locus; routed out at Gate 1 to the sibling protocol, where its own walk admits the sibling verdict. The router is clean in both directions. *Goldbach and the twin primes:* Gate 1 branch two, since no conjecture of that shape resides on any symmetry's fixed set, so no native symmetry exists and the candidates continue; they fail at Gate 3 for want of a theorem-grade wall record and rest at the ordinary openness, which is the honest state for evidential openness without walls. This control is also what keeps branch two honest: it is the branch the flagship takes, and a decisive branch whose only occupant is the thing the protocol was written for is a branch that has never been tested, so the control that enters it and **fails downstream** is the negative control the branch requires. *Chart-artifact questions of the strip-width class:* they dissolve under the Gate 2 strip with no determinate residue and are retyped rather than answered. The controls are what keep the verdict rare and the economy three-valued.

Appendix D · The Master Barrier Ledger: Three Classes, Fifteen Rows

The complete record in one place, sorted by what each row's quantifier ranges over. This is the table Section 5 derives and Section 10.2 turns into a criterion. **Class I is never citable as the total wall**, by Row II.8, and that bar governs every use of this table including this one.

Table 7 | The master barrier ledger. Fifteen rows in three classes, enumerated on seventeen lines: Row I.5 is one row with three wings, a, b, and c, listed separately because their warrants differ. The sorting axis is the quantifier, not the strength.

Row	Barrier	Quantifies over	Authority	Lag	Warrant
CLASS I · TECHNIQUE-INDEXED	*arrives after the family it blocks; complete over the born, silent over the unborn*				
I.1	Relativization	relativizing arguments	Baker-Gill-Solovay 1975	**10 yr**	Type T, unconditional
I.2	Natural proofs	constructive-and-large strategies	Razborov-Rudich 1994; JCSS 1997	**9 yr**	Type C, on the crypto hypothesis
I.3	Algebrization	algebraic-oracle arithmetization	Aaronson-Wigderson 2008	**18 yr**	Type T
I.4	Locality	local reductions and magnification	CHOPRS 2020; JACM 2022	**2 yr**	Type T within scope
I.5a	Occurrence obstructions	GCT occurrence obstructions	Bürgisser-Ikenmeyer-Panova 2016; JAMS 2019	**15 yr**	Type T
I.5b	Rank methods	rank-based lower bounds	Efremenko-Garg-Oliveira-Wigderson 2018	not dated	Type T within scope
I.5c	Algebraic natural proofs	the natural shape, algebraic register	Forbes-Shpilka-Volk; GKSS 2017; Theory of Computing 2018	not dated	Type C
I.6	Bounded-arithmetic unprovability	named fragments of the ladder	Razborov 1995	ladder-read	Type C
CLASS II · TYPE-INDEXED	*zero lag, zero foresight; binds the unborn identically with the born*				
II.1	The fixed-point-free involution	any residence argument's target set	§4.3; classical	**none**	Type T on the eigenstructure
II.2	Orientation-blindness	any rotation-invariant functional	§4.5; Weyl	**none**	Type T, an identity
II.3	The physical bar	any physical or thermodynamic route	§§7–8; MI 7.19e-05	**none**	Type T at floor and arrow
II.4	The misattribution bar	any wall attribution	§5.3; 0 bits vs 1 bit	**none**	engineering / structural
II.5	The metatheoretic rung law	every derivation-bearing standpoint	Turing 1939; Feferman 1962	**none**	Type T per rung
II.6	The access structure	all eight verdict-side actions	§5.4.2	**none**	operational / engineering
II.7	The jurisdiction type-check	one lane, one direction	§5.4	**none**	Type T on the type-check
II.8	The soundness cell	one closing argument	§5.4	**none**	Type T on the cell
CLASS III · CLOSURE	*a fact about the row-space, not a row of the object classes*				
III.1	The closure barrier	the catalogue's own totality	§5.5; Gödel, Tarski, the eigenstructure	**none**	Type T on the fixed-point-freeness

How to read the lag column. It is the paper's sharpest single artifact. Class I's lags are 10, 9, 18, 15, and 2 years: each barrier waited for its technique family to be described before it could be proven, which is what it means to quantify over descriptions. Class II's column reads **none** on every row, and that is not an absence of data. It is the class's defining property: a barrier that quantifies over the structure any technique must engage has nothing to wait for. It binds at utterance, and it bound the question before anyone asked it, because a structural fact has no publication date.

The generator. A formalized method class is an invariance signature; the signature is a blindness; the blindness is the wall. Classes and walls are born paired, which is why Class I keeps arriving and why it can never close: total coverage of every method class would be independence, and independence is a theorem nobody has. *Corroboration-grade on five for five; the universal law is unproven and is stated as unproven.*

The retired generator. Mutual information between *presupposes the foundation* and *walled*, transcribed from the recorded arc battery over a contingency table not printed here: **0 bits**, and it is trivial rather than informative if the foundation predicate is constant across the record, which is the reading §5.3's *beneath everything* invites. Between *carries an invariance signature* and *walled*: exactly **1 bit**. Foundational presence explains no wall. The walls are invariance-indexed. Permanently retired.

Row III.1 binds this table first. No claim here says the record is finished. The catalogue is completable relative to any stage and finalizable at none.

Appendix E · The Standalone Reproduction Protocol

This protocol lets an independent reader, human or machine, reproduce the paper's verdicts from the primitives without consulting the paper's conclusions. It is stated as an ordered procedure. **It is not a proof procedure and it certifies nothing about P versus NP.** It is a check on derivational soundness and on freedom from hidden steps, and its honest reach is exactly that: it verifies that the conclusions follow from the premises and the cited results, and it is not a substitute for the external warrant those results carry on their own authority.

An earlier form of this protocol carried no terrain computation and no terminal branch, and a faithful execution of it therefore could not return the verdict of Section 9.3. That defect is repaired here. The steps below are the executable form of the whole paper.

Step 0 · Intake. Parse the candidate. Refuse a formulation whose vocabulary denotes no arithmetic sentence; admit the arithmetic residue. Reject a proposition that embeds the procedure's own verdict as its content. *Returns:* the admitted string, or a refusal with the failing vocabulary named.

Step 1 · The strip. Inventory the interpretive shells. For each, vary the chart with the structure held fixed and record whether the reading changes; a reading that varies with the chart alone is gauge and is stripped. Run the neighbour test on the innermost joint: read the same shape at adjacent registers and record whether the answer is constant. *Returns:* the shell count, the residue, and the joint's classification as gauge or load-bearing. *Expected for this paper:* five shells, a Π^0_2 residue, joint load-bearing per Table 2.

Step 0 · The certification. Before any step below, run the single-seed battery of Appendix H at seed 20260622 and compare its digest against the recorded value. A matching digest certifies every computed number in the paper in one act and makes Steps 1 through 7 confirmations rather than discoveries. A value drift beyond the stated tolerances falsifies the corresponding claim; a byte-level digest drift with every value intact indicates numerical-environment variance and falsifies nothing, per Appendix H's environment clause.

Step 2 · The form. Apply §4.3's criterion: determine whether the conjecture is a residence claim on the fixed set of a symmetry specifiable without its target set. Where one exists it is native; where none does, record that and identify the symmetry the question most naturally suggests instead. Compute its eigenstructure at machine precision on the centered content space. Record eigenvalues, the involution residual, and the fixed-subspace dimension. **Route on the fixed dimension:** zero continues here; one exits to the sibling protocol. *Returns:* the eigenstructure and the route. *Expected:* all eigenvalues −1, residual 0.0, dimension 0, continue.

Step 3 · The string. Classify the quantifier register of the residue in both directions. Type the determinacy and state its cap. *Returns:* the classification and the cap. *Expected:* Π^0_2 separation, Σ^0_2 collapse, determinacy premise-capped at $\mathbb{N}$-truth.

Step 4 · The walls. Assemble the record. For each candidate wall, record the theorem, the citation, the scope, and whether it quantifies over techniques or over structure. Compute the arrival lag of each technique-indexed row against its technique family. Record the invariance signature of each walled method family and confirm the five-generation pairing. The misattribution figures of 0 bits and 1 bit are transcribed in this paper over a contingency table it does not print, so this step does not ask for them and a reader should not attempt to reproduce them from anything printed here. *Returns:* the record sorted by class, the lag vector. *Expected:* fifteen rows in three classes; lags 10, 9, 18, 15, 2.

Step 5 · The bridge. Identify the single premise separating the physical face from the class separation. Test it against the completeness theorem. *Returns:* the premise and its logical relation to the conjecture. *Expected:* logically identical, by Theorem 1.

Step 6 · The ranks. Encode the three streams over N contexts. Compute the encoding rank as the number of strictly positive Gram eigenvalues. Compute the discrimination rank against each target separately as the number of rows whose likelihood ratio for that target exceeds one. **Compute them separately and do not conflate them.** *Returns:* a rank pair per target. *Expected:* against the physical face, (3, 3); against the string, (3, 1); against the halt-proposition, (3, 2), taking §6.3's seal on the conditional at the declared premise cap.

Step 7 · The reliability layer. For any determinant carrying a verdict: compute it by four independent estimators and confirm the spread is within the conditioning-scaled tolerance $4{\cdot}\kappa(R){\cdot}\varepsilon$; confirm the identity $|\lambda^2 - \det(R)|$ within the same tolerance; escalate to fifty-digit precision if either fails or if a verdict margin sits inside the tolerance band. A verdict that cannot be made reliable is emitted engineering-incomplete and never forced. *Returns:* the spread, the residual, the margins in orders, and the escalation flag.

Step 8 · The aperture. Check for a supplied witness. If present, adjudicate by direction and stop. If absent, locate the aperture, type it supply-only, enumerate the exits, and emit one sentence.

Step 9 · The gates. Run the five gates of Appendix C against the assembled record. Confirm the four riders. *Returns:* the verdict or the routing.

Step 10 · The emission. Emit the composite as a tuple. **Do not compress it.** Attach the warrant grade to every face. Mark unreached stages as not reached and fabricate no trace.

What a faithful execution returns. The physical face sealed and answer-independent; the geometric shape sealed on the eigenstructure; the string terminal in the terrain; the fence on the named inflations; the coincidence claim broken for absence of warrant. **What it does not return, and cannot:** any verdict on P versus NP itself. A reader whose execution returns one has made an error, and Step 2 plus Step 6 are where to look for it.

Appendix F · Executable Exhibits and the Recorded Battery

The exhibits below are the sections of the single-seed certification battery of Appendix H, which executes them together with the involution facts of Section 4.2 as one script, in one pass, at seed 20260622 in double precision, and records one digest over the whole. They produce every numeric claim in this paper **except three**, and the three exceptions are named rather than left for a reader to find, here and again in Appendix H, whose certificate is scoped to exclude them. Failure of any check on re-execution falsifies the corresponding claim.

The exceptions, and there are three. The mutual-information estimate of 7.19×10^{-5} is transcribed from the recorded arc battery and is not recomputed here; §9.1 states its status, and the value it corroborates, exactly zero, is proved at §9.3 rather than measured. The **misattribution figures of 0 bits and 1 bit** at §5.3, Row II.4, and Appendix D are likewise transcribed, and the contingency table they are computed over is **not printed in this paper**, so Appendix E Step 4 cannot ask a reader to reproduce them and no longer does. Their constraints are worth stating, because they are tight and a reader is entitled to test them: 1 bit exactly between two binary variables forces $H(Y) = 1$ and $H(Y|X) = 0$, which is the record exactly half walled with the signature bijective on it, an even census with an exact split; and 0 bits is trivial rather than informative if *presupposes the foundation* is constant across the record, which §5.3's own phrase *beneath everything* implies. **Row II.4's law does not rest on those two numbers.** It rests on the five-generation invariance record, which is printed, and which is what a reader should attack. And the three rank pairs, three-three against the physical face, three-one against the string, and three-two against the halt-proposition, are **argued and not computed**: §9.3 establishes the discrimination rank by an invariance argument over the two truth values of the separation and the encoding rank by the observation that linear independence is unaffected by co-variation with an external indicator. Neither is read off a $\det(R)$ computed on this paper's own encoding, and no such encoding is printed anywhere in this paper. Anyone reporting those ranks as computed, this paper's own appendices included, is reporting them wrongly.

PYTHON

```python
import numpy as np

# EXHIBIT A - complementation on the four-string universe
U = ['00','01','10','11']
langs = [frozenset(s for j,s in enumerate(U) if (i>>j)&1)
for i in range(16)]
comp = lambda L: frozenset(set(U)-set(L))
fixed = [L for L in langs if comp(L)==L]
seen, cycles = set(), []
for L in langs:
    if L in seen: continue
    seen |= {L, comp(L)}; cycles.append((L, comp(L)))
assert all(comp(comp(L))==L for L in langs)      # the
involution law, CHECKED not asserted
# -> languages=16  two-cycles=8  fixed points=0
involution law holds for all 16

# EXHIBIT A2 - the counter-exhibit of Section 4.4, run
against ourselves
flip  = lambda L: frozenset(x[0] + ('1' if x[1]=='0' else
'0') for x in L)
sig_p = lambda L: comp(flip(L))                  # a
composite, the freedom RH's side is granted
assert all(sig_p(sig_p(L))==L for L in langs)    # still
an involution
fix_p = [L for L in langs if sig_p(L)==L]
# -> Fix(comp)   = 0  languages
# -> Fix(sigma') = 4  languages: {00,10} {01,10} {00,11}
{01,11}
# -> so fixed-point-freeness is a property of the MAP,
not of the terrain.
#    Section 4.4 retires the terrain claim. The criterion
survives: none of these
#    fixed sets is what P vs NP resides on, which is P.

# EXHIBIT B - eigenstructure of the two involutions
# NOTE: the bare functional equation s -> 1-s fixes only
the POINT 1/2 and is -I,
# fixed dim 0, the SAME signature as complementation. The
critical line is the fixed
```

```
# set of the COMPOSITE s -> 1-conj(s) (functional
equation . Schwarz reflection).
# Section 4.2. The 4x4 forms below are the kernel's
centered reading of each.
delta = -np.eye(4)                    # complementation
on languages, and s -> 1-s
sigma = np.diag([1.,-1.,-1.,-1.])     # s -> 1-conj(s),
the composite that fixes the line
for M in (delta, sigma):
    ev  = np.linalg.eigvalsh(M)
    res = np.abs(M@M - np.eye(4)).max()
    dim = int(np.sum(np.isclose(ev, 1.0)))
# delta -> eigenvalues [-1,-1,-1,-1]  residual 0.0  fixed
dim 0
# sigma -> eigenvalues [-1,-1,-1,+1]  residual 0.0  fixed
dim 1

# EXHIBIT C - orientation-blindness of the squared
invariant
def qmul(a,b):
    w1,x1,y1,z1=a; w2,x2,y2,z2=b
    return np.array([w1*w2-x1*x2-y1*y2-z1*z2,
w1*x2+x1*w2+y1*z2-z1*y2,
                     w1*y2-x1*z2+y1*w2+z1*x2,
w1*z2+x1*y2-y1*x2+z1*w2])
rg = np.random.default_rng(20260622); N = 24
M = rg.standard_normal((3,N)); M -=
M.mean(1,keepdims=True)
M /= M.std(1,ddof=1,keepdims=True)
Q = M/np.sqrt((M*M).sum(1,keepdims=True))
B = np.linalg.svd(Q, full_matrices=False)[2][:3]     #
basis fixed ONCE, reused after reflection
def lam_det(Qm):
    co = Qm@B.T; q = [np.concatenate(([0.],c)) for c in
co]
    return float(qmul(qmul(q[0],q[1]),q[2])[0]),
float(np.linalg.det(Qm@Qm.T))
l0,d0 = lam_det(Q)
l1,d1 = lam_det(np.diag([-1.,1.,1.])@Q)              #
reflect one axis = negate the claim
# baseline   lambda = -0.965027450462   det(R) =
0.931277980144
# reflected  lambda = +0.965027450462   det(R) =
0.931277980144
# |det(R)_P - det(R)_notP| = 0.0 exactly     lambda ratio
= -1.000000 exactly
# |lambda^2 - det(R)| = 5.551e-16           kappa(R) =
1.6568

# EXHIBIT D - certify-completely has no fixed point
chain = [10]
for _ in range(5):
    chain.append(chain[-1] + 1)   # each total
certificate spawns exactly one row it cannot contain
# -> [10, 11, 12, 13, 14, 15]  strictly increasing  no
fixed point  (no RNG consumed)
swallow = -np.eye(4)              # total self-inclusion:
the container inside its contents
# -> eigenvalues [-1,-1,-1,-1]  residual 0.0  fixed dim
0  == the delta-root signature
```

Recorded results. *Exhibit A:* 16 languages, 8 two-cycles, 0 fixed points, involution law asserted by `assert` and confirmed for all 16. *Exhibit A2, the counter-exhibit:* Fix(comp) = 0 languages against Fix(σ') = 4 languages, namely {00,10}, {01,10}, {00,11}, {01,11}, with σ' still an involution. This is Section 4.4 executing against the paper's own earlier claim: fixed-point-freeness is a property of the map and not of the terrain, and the terrain claim is retired. What survives is the criterion, since none of those four fixed sets is what P versus NP resides on, which by Theorem 1 is P. *Exhibit B:* the two eigenstructures as printed, δ at fixed dim 0 and σ at fixed dim 1, δ being both complementation and the bare functional equation $s \mapsto 1 - s$, σ being the composite $s \mapsto 1 - \bar{s}$ that fixes the critical line; the gap is 1 against 0 and Section 4.4 is why it is the exhibit rather than the reason. *Exhibit C:* run on **synthetic Gaussian input**, since the identity it demonstrates holds for arbitrary input and noise is therefore the honest witness that it is not fitted; the determinant is bit-identical under negation at a difference of exactly zero, while the signed scalar flips at a ratio of exactly −1.000000; the identity closes at 5.551×10^{-16} against a conditioning-scaled tolerance of $4{\cdot}\kappa{\cdot}\varepsilon = 1.47 \times 10^{-15}$, where $\varepsilon = 2^{-52} = 2.220446 \times 10^{-16}$ is machine epsilon and not the unit roundoff $u = 2^{-53}$, the convention stated because the two differ by a factor of two and the earlier form of this paper wrote u while computing with ε, and $\kappa(R) = 1.6568$ sits far inside the gate. These numbers certify the identity and certify nothing about P versus NP. *Exhibit D:* the chain is strictly increasing, increments by exactly one per Row III.1, consumes no RNG and carries no seed, and the self-inclusion map carries the identical signature as complementation, one engine at both ends of the stack.

Not reached, and not traced. The mutual-information estimate of 7.19×10^{-5} reported at Section 9.1 is transcribed from the recorded arc battery and is not recomputed here; it is an estimator output at the declared seed and carries engineering grade. No stage this paper did not reach is traced anywhere in it.

Appendix G · Author's Provenance and Method Disclosure

This paper was developed under Trisduction, a verification and organizing discipline, not a source of results. Its conclusions rest solely on the standard results cited in the body. Trisduction is the method under which those results were decomposed, assembled, and audited. It adds them no warrant and claims no authorship over them.

Trisduction was used for fidelity. It forces a claim onto three independent axes so no single persuasive line carries it alone, requires every verdict to resolve to one of three states, sealed, broken, or open, each with a named failure mechanism, and attaches an explicit warrant grade, theorem, conditional, structural, or premise, to every claim so nothing is stated above its strength. Two errors it is built to catch are inflation, reading an internal lock as a proof, and circularity, reading a restatement of a claim as a derivation of it.

The root axiom, RA, is that to exist is to actuate: every physical existent carries a strictly positive energetic cost, grounded in the Heisenberg energy floor, the zero-point energy, and Landauer's bound. The formal root inside it is that to formally be is to be grounded, with provability and computation levels of access to a determinacy fixed at the ground rather than ingredients of it.

The key procedures, in brief. Orthogonal triaxial convergence: a proposition is split into three disjoint axes, formal-structural, empirical or dynamical, and registrational, verified by three separate instrument sets, agreement across the three the operational meaning of a seal. The Geometric Orthogonal Lock, GOL: the three axes are read as vectors and their independence is tested by the determinant of their correlation matrix, a determinant clear of zero a genuine three-dimensional lock, a collapse to zero one axis dissolving into the plane of the other two, a broken lock. The Convergence Dissolution Test, CDT: before the lock is read, any mass-bearing common cause the three axes might share is projected out, so an apparent convergence that traces to a shared source rather than to independent roads dissolves and carries no warrant, the lock computed on the residue that survives. The twelve-gate cascade: twelve directed failure screens, self-reference, frame-dependence, missing mechanism, and the rest, run in order, the first failure terminating the verdict with its mechanism named. The verdict issues in the three-state economy with its warrant grade attached, and where the argument stops short the open direction is named rather than filled.

Box G1 | The GOL kernel identity, proved

The lock is a determinant identity, not a metaphor. Let the three axes, after normalization, be unit vectors expressed in an orthonormal basis of the subspace they span and read as pure quaternions a, b, c, with norm one each. Hamilton's product of two pure quaternions is $p\,q = -(p\cdot q) + p\times q$, the real part the negative dot product and the imaginary part the cross product, checked on the units by $i\,j = k = i\times j$ and $i\,i = -1 = -(i\cdot i)$. The lock scalar is the real part of the triple product, $\lambda = \mathrm{Re}(a\,b\,c)$.

Expanding, $a\,b = -(a\cdot b) + a\times b$, so $\mathrm{Re}(a\,b\,c) = -(a\cdot b)\,\mathrm{Re}(c) + \mathrm{Re}((a\times b)\,c)$. The first term is zero since c is pure, and the second is $-(a\times b)\cdot c$ by the same product rule, giving

$\lambda = -(a\times b)\cdot c = -\det[a\ b\ c]$,

minus the signed volume of the parallelepiped the three axes span. Writing A for the matrix whose columns are a, b, c, the correlation matrix is $R = A^{T}A$, its entries the pairwise dot products, so

$\det(R) = \det(A^{T}A) = \det(A)^2 = \lambda^2$.

The kernel identity $\lambda^2 = \det(R)$ is that the squared signed volume equals the Gram determinant, with the quaternion triple product the machine that computes the volume. It is confirmed at machine precision, the residual $|\lambda^2 - \det(R)|$ at 2×10^{-16} on the method's own master battery. Appendix F of this paper runs a different construction and returns 5.551×10^{-16} for the same identity; the two are separate batteries at separate inputs and neither is the other's figure.

Four consequences fix the reading, and they are why the number can be trusted to say what the lock says. The determinant is bounded, det(R) in the interval from zero to one for unit axes, by Hadamard's inequality above and positive-semidefiniteness below, so the lock has a hard ceiling at one, the fully orthogonal frame, and a hard floor at zero. The floor is the break: det(R) equals zero exactly when the three axes are linearly dependent, one axis lying in the plane of the other two, which is the geometric content of a collapsed lock. The magnitude is orientation-blind: reflecting any axis sends A to a matrix of opposite determinant, flipping the sign of λ while det(R) equals λ^2 is unchanged, so the number certifies the dimensionality of the lock and never its truth-sign, which is read from the ordered axes and not from the scalar. And the magnitude is frame-invariant: rotating all three axes together is conjugation q to $u\,q\,\bar{u}$ on the imaginary quaternions, under which the real part is preserved and dot and cross products are covariant, so λ and det(R) depend on the configuration and not on the coordinate labels.

The axis count is three because the algebra forces it, not because three was chosen. The audit-composition law requires associativity, so iterated audits bracket the same way, and the absence of zero divisors, so nonzero warrants never compound to nothing. By Frobenius's theorem the only finite-dimensional associative division algebras over the reals are the reals, the complex numbers, and the quaternions, and three mutually orthogonal imaginary axes exist only in the quaternions, whose imaginary units i, j, k are the three axes. The next normed division algebra, the octonions with seven imaginary units, fails associativity, witnessed by the nonzero associator of e_1, e_2, e_4, so it cannot carry the composition law and the construction stops at three. The three axes are the imaginary part of the unique associative real division algebra that supports the audit.

This paper in particular. The three axes were the structural row, NP-completeness by Cook-Levin with the restricted lower bounds of Håstad and Smolensky and the fifteen-row barrier record of Section 5; the physical row, the actualized generate-verify asymmetry charged by the nonequilibrium work relations, bounded in rate by the quantum speed limits and in parallel extent by the distinguishable-state count; and the registrational row, the check-cheap generate-costly verifier architecture read in the statistical zero-knowledge regime, chosen over the computational regime because the computational guarantee would presuppose the hardness under audit and sink the row on circularity. The lock is read three times against three different targets and the readings are the paper's cleanest instrument: (3,3) against the physical face, (3,1) against the string, and (3,2) against the halt-proposition, the last taking §6.3's seal on the conditional at the declared premise cap. Against the physical face the encoding rank is three and the discrimination rank is three, and the face seals. Against the stripped arithmetic string the encoding rank is still three while the discrimination rank falls to one, by the truth-invariance computation of Section 9.3: the physical and registrational rows take identical values under both truth values of the separation and carry zero mutual information with it, the estimator's noise floor of 7.19×10^{-5} recorded at Section 9.1 and load-bearing on nothing, so the only row that shifts is the admitted premise, and the operator declines to adjudicate a claim on warrant equivalent to the claim and routes the string out of itself to the five-gate protocol of Appendix C. **The lock is not computed on this paper's own encoding, and no reader should think it is.** The ranks quoted above are argued and not computed: §9.3 establishes the discrimination rank by an invariance argument over the two truth values of the separation, and the encoding rank by the observation that linear independence is unaffected by co-variation with an external indicator. No $\det(R)$ is computed on the paper's warrant streams anywhere in the paper, and no such encoding is printed. Exhibit C of Appendix F, which returns $\lambda = -0.965027450462$ and $\det(R) = 0.931277980144$ with the identity closing at $|\lambda^2 - \det(R)| = 5.551 \times 10^{-16}$ at $\kappa(R) = 1.6568$, runs on **synthetic Gaussian input at seed 20260622 over N = 24 contexts**, and that is the correct input for what it demonstrates: the orientation-blindness of §4.5 is an identity that holds for *arbitrary* input, so exhibiting it on noise is what shows it is not fitted. Under negation of the claim the determinant is bit-identical at a difference of exactly zero while the signed scalar flips at a ratio of exactly −1.000000. Those numbers certify the identity and they certify nothing about P versus NP, and they are sections C3 and C4 of the single-seed certification of Appendix H, whose one digest covers them together with every other computed number in the paper. Exhibit B carries Face Two and is not synthetic: complementation is fixed-point-free with every eigenvalue −1, residual 0.0 exactly, fixed-subspace dimension zero, against dimension one for the composite $s \mapsto 1 - \bar{s}$ that fixes the critical line. The load-bearing step is Theorem 1 of Section 7: the single premise standing between any physical resource bound and the class separation is logically identical to the conjecture by Cook-Levin, which is what collapses the discrimination rank to one, and Theorem 2 of Section 8 is what shows that premise admits no physical discharge. The verdict is a three-face composite, not one token: Face One, the kinetic field, sealed at theorem grade on the cost floor and the thermodynamic arrow and scope-fenced from the string by the proved mutual-information floor of exactly zero; Face Two, the geometric shape, sealed at theorem grade on the eigenstructure and silent on the truth-sign by identity; Face Three, the formal string, [$\varnothing_0$] the grounded-sealed halt, **structural grade on Type-T legs** and not theorem grade, with its determinacy held at exactly the stated premise cap, terminal against method, reframing, and instrument, and barred by its own closure row from any claim of permanence, the halt sealed and a proof of the string neither offered nor claimed; and beside the faces one broken verdict scoped to named inflations only, including the inflation that the fixed-point structure forces the separation as a proof, which the paper retires against itself on the Immerman-Szelepcsényi record. No new theorem is claimed anywhere. Every leg is classical and cited, the arrangement is the whole contribution, and $\Delta M = 0$.

The canonical statement of the method, its axioms, and its executable batteries lives in the reference cited below, continuously updated at the same location. This note is a pointer, not a substitute.

Appendix H · The Single-Seed Certification

The law. Every computed number in this paper is produced by one battery, `certify.py`, executed in one pass at one seed, 20260622, in double precision. The battery's sections are the exhibits of Appendix F together with the involution facts of Section 4.2, which the prose prints and the certification computes: C1 the two symmetries of the zero set and the three zero checks, C2 the language-space exhibits including the counter-exhibit, C3 the four-dimensional eigenstructures, C4 the kernel and the orientation-blindness identity with the Cauchy-Binet corroboration of Section 6.1, C5 the closure chain and the self-inclusion signature. The battery emits a canonical transcript and one SHA-256 digest over it. The digest is the paper's numeric identity: one number that changes if any certified number changes.

What the certificate is, and what it is not. It is a reproducibility certificate at engineering grade: anyone re-executing the battery must reproduce every value below at the stated precision, and the transcript byte for byte where the numerical environment matches. It is not a truth certificate for any theorem, and it is not evidence about P versus NP: section C4 runs on synthetic Gaussian input precisely because the identity it exhibits holds for arbitrary input, so noise is the witness that nothing is fitted. The certificate binds the paper's arithmetic to its text. The theorems are bound by their proofs and the assembly by its citations, neither of which a hash can carry.

The seed-sensitivity ledger. Exactly one section consumes the random generator, C4. Sections C1, C2, C3, and C5 are RNG-free and seed-inert, their values identical under any seed, and that asymmetry is itself part of the certificate: only one number family in this paper is stochastic at all, and it is the one whose point is that its values do not matter.

The three exclusions, named so the certificate cannot be read as wider than it is. The estimator noise floor of 7.19×10^{-5} is transcribed from the recorded arc battery and is not recomputed here; the certified value on that question is exactly zero, proved at Section 9.3. The misattribution figures of 0 bits and 1 bit are transcribed over a contingency table this paper does not print, and Row II.4's law rests on the five-generation record instead. The three rank pairs, (3, 3), (3, 1), and (3, 2), are argued and not computed, and no $\det(R)$ is computed on this paper's own encoding anywhere. A certificate that silently absorbed these would be the traceability defect of this paper's own audit history reintroduced at the moment of unification, and it is refused.

The falsification clause. Failure of any check on re-execution falsifies the corresponding claim in the body, and any such failure changes the digest. A byte-level digest drift with every value intact at the stated precision indicates numerical-environment variance in the last units of the floating-point results and falsifies nothing; the values, not the bytes, carry the claims, and the digest is recorded on the build environment stated in Appendix E.

The recorded transcript, verbatim.

```
SINGLE-SEED CERTIFICATION BATTERY : seed 20260622 :
double precision
eps = 2^-52 = 2.220446e-16 (machine epsilon; the unit
roundoff u = 2^-53 is eps/2)

C1 : the two symmetries of the Riemann zero set, centred
at 1/2 on the plane
  s -> 1-s       eigenvalues [-1, -1]  involution
residual 0.0  fixed dim 0
  s -> 1-conj(s) eigenvalues [-1, +1]  involution
residual 0.0  fixed dim 1
  zero t=14.134725142 : |rho-(1-rho)| = 28.269450 : |rho-
(1-conj(rho))| = 0.0
  zero t=21.022039639 : |rho-(1-rho)| = 42.044079 : |rho-
(1-conj(rho))| = 0.0
  zero t=25.010857580 : |rho-(1-rho)| = 50.021715 : |rho-
(1-conj(rho))| = 0.0
  the functional equation alone fixes the point 1/2 (dim
0, complementation signature);
  the composite with Schwarz reflection fixes the
critical line (dim 1). Section 4.2.

C2 : complementation and the counter-exhibit on the four-
string universe
  comp    : languages=16  two-cycles=8  fixed points=0
involution law all 16
  sigma'  : involution confirmed  Fix(sigma') = 4
languages : {00,10} {00,11} {01,10} {01,11}
  fixed-point-freeness is a property of the map, not the
terrain. Section 4.4.

C3 : eigenstructure of the two involutions on the four-
dimensional content space
  delta : eigenvalues [-1, -1, -1, -1]  residual 0.0
fixed dim 0
  sigma : eigenvalues [-1, -1, -1, +1]  residual 0.0
fixed dim 1

C4 : orientation-blindness of the squared invariant : the
ONE stochastic section
  baseline   lambda = -0.965027450462   det(R) =
0.931277980144
  reflected  lambda = +0.965027450462   det(R) =
0.931277980144
  |det(R)_P - det(R)_notP| = 0.0 exactly   lambda ratio =
-1.000000 exactly
  |lambda^2 - det(R)| = 5.551e-16   kappa(R) = 1.6568
tolerance 4*kappa*eps = 1.4716e-15
  Cauchy-Binet : det(QQ^T) - sum of squared 3x3 minors =
7.772e-16 (Section 6.1)
  synthetic Gaussian input by design: the identity holds
for arbitrary input,
  so noise is the witness that nothing is fitted.
Certifies the identity,
  certifies nothing about P versus NP. Appendix G.

C5 : certify-completely has no fixed point
  chain = [10, 11, 12, 13, 14, 15]  strictly increasing =
True  no fixed point  (no RNG consumed)
  swallow : eigenvalues [-1, -1, -1, -1]  residual 0.0
fixed dim 0  == the delta signature

SEED-SENSITIVITY LEDGER : C4 consumes the generator; C1,
C2, C3, C5 are RNG-free
and seed-inert, their values identical under any seed,
which is part of the
certificate: only one number family in this paper is
stochastic at all.

EXCLUSIONS, three, outside the span of this battery and
typed where they live:
  1. 7.19e-05, the estimator noise floor, transcribed
from the recorded arc
     battery; the certified value is exactly zero, proved
at Section 9.3.
  2. the misattribution figures 0 bits and 1 bit,
transcribed over a
     contingency table this paper does not print; Row II.
4 rests on the
     five-generation record, not on them.
  3. the three rank pairs (3,3), (3,1), (3,2), argued and
not computed;
```

```
     no det(R) is computed on this paper own encoding
anywhere.
```

The recorded digest.

```
SHA-256(transcript), line-wrapped for the page and read
as one 64-hex string:
    de607729f577bddedb392f57cd090804
    0b2502f6a1f4807ba166bdbf8a27f27d
```

One seed, one pass, one digest. The paper's numbers are one object, and this is its name.